Casting_Inspection_Industrial_Workflow

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1 Automated Casting Quality Inspection Workflow (Prepared by Nay Linn Aung)

Industrial computer vision reference notebook for cast aluminum part inspection

Primary dataset: Kaggle **Casting Product Image Data for Quality Inspection**

Target architecture: **edge-first ResNet classification gate**, with a controlled extension path to **Ultralytics YOLO detection, segmentation, and oriented bounding boxes** when localized annotations become available.

It is a production-pilot reference template with explicit controls for reproducibility, dataset integrity, threshold governance, error analysis, export validation, drift monitoring, and release readiness.

1.1 1. Executive Overview

1.1.1 Problem Statement

Cast aluminum components in automotive manufacturing require repeatable visual inspection for defects such as porosity, blow holes, shrinkage, burrs, inclusions, deformation, surface discontinuities, and other process-related irregularities. Manual inspection varies across operators, shifts, fixtures, camera stations, suppliers, and plants.

The first deployable objective is a **binary part-level inspection gate**:

- **OK**: no visible defect evidence requiring rejection.
- **Defective**: visible defect evidence requiring rejection, rework, or quality engineering review.

The operational priority is minimizing **false accepts** while keeping false rejects within a manageable review/rework load.

1.1.2 Why This Dataset

The public casting dataset is useful for a pilot template because it contains real industrial casting imagery with image-level labels and a stable folder layout:

```
casting_data/casting_data/
  train/
    def_front/
    ok_front/
  test/
```

```
def_front/
ok_front/
```

It is appropriate for a classifier baseline. It is not, by itself, sufficient for production release or localized defect modeling.

1.1.3 Why This Stack

The inspection architecture is intentionally modular:

- **OpenCV** for deterministic preprocessing, image-quality diagnostics, and edge guardrails.
- **ResNet-family classifier** for fast image-level accept/reject decisions from existing labels.
- **Ultralytics YOLO detection** for localized defect boxes once annotations exist.
- **Ultralytics YOLO segmentation** for defect-area and severity measurement once masks exist.
- **Ultralytics YOLO OBB** for rotated geometry where axis-aligned boxes are too imprecise.

1.1.4 Scope and Non-Goals

Covered: data validation, leakage checks, preprocessing, augmentation, transfer learning, advanced evaluation, threshold policy, error review, Grad-CAM diagnostics, artifact/export controls, edge latency, drift handling, risks, acceptance criteria, and a model-card summary.

Not covered: PLC/MES integration, certified plant acceptance testing, multi-camera calibration, final defect taxonomy approval, or commercial release based only on public data.

1.2 2. Assumptions & Constraints

1.2.1 Data Assumptions

- Dataset is available locally, in Kaggle, or through `CASTING_DATA_ROOT`.
- Labels are image-level binary labels only.
- Public data may include acquisition artifacts, pre-applied augmentation, repeated visual patterns, and distribution shift relative to plant imagery.
- Production release requires plant-owned or properly licensed validation data from the actual station.

1.2.2 Hardware Assumptions

The intended runtime is edge-first:

- Industrial PC, CPU-only IPC, NVIDIA Jetson-class device, or GPU-enabled edge workstation.
- Batch size 1 inference.
- Deterministic preprocessing.
- Local fail-safe handling for camera faults, unreadable frames, inference failures, and downstream I/O issues.
- Latency measured across the full station path, not only model inference.

1.2.3 Labeling Limitations

Current labels do not identify defect type, location, count, severity, rework disposition, or visibility from another camera angle. Localization work must begin with new annotations, not assumptions inferred from image-level labels.

1.2.4 License and Provenance Constraint

The public dataset is suitable for this reference workflow and pilot exploration. Commercial deployment requires license, provenance, privacy, and supplier-data review.

[1]:
```
# Code Intent:
# Establish a reliable runtime before any dataset or model work begins.
# Key Ideas:
# - Separate mandatory dependencies from optional integrations so the notebook␣
↪fails only on true blockers.
# - Record package, Python, platform, CUDA, and seed details for␣
↪reproducibility and audit review.
# - Force deterministic PyTorch behavior where practical so repeated pilot runs␣
↪are comparable.
# - Select CPU/GPU automatically while making the selected device visible to␣
↪reviewers.
# - Build a preflight report that later becomes part of the experiment manifest.

# Environment, versions, deterministic controls, and runtime preflight

from __future__ import annotations

import copy
import hashlib
import importlib
import importlib.metadata as importlib_metadata
import importlib.util
import json
import math
import os
import platform
import random
import subprocess
import sys
import time
import warnings
from dataclasses import asdict, dataclass
from datetime import datetime, timezone
from pathlib import Path
from typing import Dict, Iterable, List, Optional, Sequence, Tuple

REQUIRED_MODULES = {
```

```
    "cv2": "opencv-python",
    "matplotlib": "matplotlib",
    "numpy": "numpy",
    "pandas": "pandas",
    "PIL": "Pillow",
    "sklearn": "scikit-learn",
    "torch": "torch",
    "torchvision": "torchvision",
}
OPTIONAL_MODULES = {
    "onnx": "onnx",
    "onnxruntime": "onnxruntime",
    "ultralytics": "ultralytics",
    "kagglehub": "kagglehub",
}

missing = [
    pip_name
    for module_name, pip_name in REQUIRED_MODULES.items()
    if importlib.util.find_spec(module_name) is None
]
if missing:
    raise ImportError(
        "Missing required packages: "
        + ", ".join(missing)
        + ". Install them in the notebook environment before executing the␣
 ↪workflow."
    )

import cv2
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from IPython.display import display
from PIL import Image

import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import DataLoader, Dataset, WeightedRandomSampler
from torchvision import models, transforms

from sklearn.calibration import calibration_curve
from sklearn.metrics import (
    accuracy_score,
    average_precision_score,
    brier_score_loss,
```

```python
    classification_report,
    confusion_matrix,
    f1_score,
    precision_score,
    precision_recall_curve,
    recall_score,
    roc_auc_score,
    roc_curve,
)
from sklearn.model_selection import train_test_split

def parse_bool_env(name: str, default: bool = False) -> bool:
    raw = os.getenv(name)
    if raw is None:
        return default
    return raw.strip().lower() in {"1", "true", "yes", "y", "on"}

def package_version(distribution_name: str) -> str:
    try:
        return importlib_metadata.version(distribution_name)
    except importlib_metadata.PackageNotFoundError:
        return "not-installed"

PACKAGE_VERSIONS = {
    pip_name: package_version(pip_name)
    for pip_name in {**REQUIRED_MODULES, **OPTIONAL_MODULES}.values()
}

SEED = int(os.getenv("CASTING_INSPECTION_SEED", "42"))
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
torch.cuda.manual_seed_all(SEED)
torch.backends.cudnn.benchmark = False
torch.backends.cudnn.deterministic = True

DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")

preflight_report = {
    "timestamp_utc": datetime.now(timezone.utc).isoformat(),
    "python": platform.python_version(),
    "platform": platform.platform(),
    "device": str(DEVICE),
    "cuda_available": torch.cuda.is_available(),
    "seed": SEED,
    "package_versions": PACKAGE_VERSIONS,
}
```

```
display(pd.DataFrame([preflight_report]).drop(columns=["package_versions"]))
display(pd.DataFrame([PACKAGE_VERSIONS]).T.rename(columns={0: "version"}))
```

```
                      timestamp_utc   python  \
0  2026-05-01T02:21:16.393212+00:00  3.12.13

                              platform device  cuda_available  seed
0  Linux-6.6.113+-x86_64-with-glibc2.35   cuda            True    42

                     version
opencv-python      4.13.0.92
matplotlib            3.10.0
numpy                  2.0.2
pandas                 2.2.2
Pillow                11.3.0
scikit-learn           1.6.1
torch           2.10.0+cu128
torchvision     0.25.0+cu128
onnx           not-installed
onnxruntime    not-installed
ultralytics    not-installed
kagglehub              1.0.0
```

[2]:
```
# Code Intent:
# Centralize all operational knobs into typed configuration objects.
# Key Ideas:
# - Keep dataset, training, preprocessing, edge, and artifact settings separate␣
↪to avoid hidden coupling.
# - Allow important runtime switches to be controlled by environment variables␣
↪for CI/Kaggle/edge use.
# - Make Colab dataset setup robust by supporting explicit roots, recursive␣
↪discovery, and optional KaggleHub download.
# - Keep artifact writing and ONNX export disabled by default so review runs do␣
↪not create unexpected files.
# - Define stable label mappings once so metrics, thresholds, logging, and␣
↪artifacts use the same contract.
# - Store commonly documented Kaggle counts as warnings, not hard failures,␣
↪because dataset variants exist.

# Configuration dataclasses

@dataclass(frozen=True)
class DatasetConfig:
    root_env_var: str = "CASTING_DATA_ROOT"
    kaggle_dataset_slug: str = "ravirajsinh45/
↪real-life-industrial-dataset-of-casting-product"
```

```
    auto_download_dataset: bool =␣
↪parse_bool_env("AUTO_DOWNLOAD_CASTING_DATASET", True)
    install_kagglehub_if_missing: bool =␣
↪parse_bool_env("INSTALL_KAGGLEHUB_IF_MISSING", True)
    image_extensions: Tuple[str, ...] = (".jpg", ".jpeg", ".png", ".bmp", ".
↪tif", ".tiff")
    val_fraction_from_train: float = 0.15
    max_audit_images: Optional[int] = None
    max_near_duplicate_images: int = 1500
    near_duplicate_hamming_threshold: int = 4
    expected_public_counts_enabled: bool = True

@dataclass(frozen=True)
class TrainingConfig:
    image_size: int = 224
    batch_size: int = 32
    num_workers: int = 0
    epochs: int = 5
    learning_rate: float = 1e-4
    weight_decay: float = 1e-4
    use_weighted_sampler: bool = True
    use_class_weighted_loss: bool = False
    defect_positive_label: int = 1
    defect_recall_target: float = 0.95
    threshold_grid_points: int = 91
    run_one_batch_smoke_test: bool = parse_bool_env("RUN_ONE_BATCH_SMOKE_TEST",␣
↪False)

@dataclass(frozen=True)
class PreprocessConfig:
    use_clahe: bool = True
    denoise: bool = False
    clahe_clip_limit: float = 2.0
    clahe_tile_grid_size: Tuple[int, int] = (8, 8)

@dataclass(frozen=True)
class EdgeConfig:
    p95_latency_budget_ms: float = float(os.
↪getenv("EDGE_P95_LATENCY_BUDGET_MS", "50"))
    benchmark_repeats: int = 50
    benchmark_warmup: int = 10
    batch_size_one_required: bool = True

@dataclass(frozen=True)
class ArtifactConfig:
    root: Path = Path(os.getenv("CASTING_ARTIFACT_ROOT", "artifacts/
↪casting_inspection"))
```

```
    save_artifacts: bool = parse_bool_env("SAVE_CASTING_ARTIFACTS", False)
    export_onnx: bool = parse_bool_env("EXPORT_CASTING_ONNX", False)
    validate_onnx_parity: bool = parse_bool_env("VALIDATE_ONNX_PARITY", True)

DATASET_CONFIG = DatasetConfig()
TRAINING_CONFIG = TrainingConfig()
PREPROCESS_CONFIG = PreprocessConfig()
EDGE_CONFIG = EdgeConfig()
ARTIFACT_CONFIG = ArtifactConfig()

CLASS_TO_INDEX = {"ok_front": 0, "def_front": 1}
INDEX_TO_CLASS = {value: key for key, value in CLASS_TO_INDEX.items()}
DISPLAY_NAMES = {0: "OK", 1: "Defective"}
DEFECT_CLASSES = {0: "casting_defect"}

PREPROCESS_VERSION = os.getenv("CASTING_PREPROCESS_VERSION", "opencv-clahe-v2")
MODEL_VERSION = os.getenv("CASTING_MODEL_VERSION",␣
 ↪"resnet18-imagelevel-public-pilot-v1")
NOTEBOOK_RELEASE_STATUS = "pilot_reference_not_production_approved"
LEAKAGE_POLICY = {
    "priority_order": ["train", "val", "test"],
    "train_val_overlap_action": "remove_from_validation",
    "test_overlap_action": "remove_from_test",
    "filename_overlap": "drop_downstream_rows_matching_upstream_filenames",
    "exact_hash_overlap": "drop_downstream_rows_matching_upstream_sha1",
    "near_duplicate_overlap": "warn_and_review_only",
}

EXPECTED_PUBLIC_COUNTS = {
    ("train", "def_front"): 3758,
    ("train", "ok_front"): 2875,
    ("test", "def_front"): 453,
    ("test", "ok_front"): 262,
}

configuration_report = {
    "workflow": {
        "model_version": MODEL_VERSION,
        "preprocess_version": PREPROCESS_VERSION,
        "notebook_release_status": NOTEBOOK_RELEASE_STATUS,
        "leakage_policy": LEAKAGE_POLICY,
    },
    "dataset": asdict(DATASET_CONFIG),
    "training": asdict(TRAINING_CONFIG),
    "preprocess": asdict(PREPROCESS_CONFIG),
    "edge": asdict(EDGE_CONFIG),
    "artifact": {**asdict(ARTIFACT_CONFIG), "root": str(ARTIFACT_CONFIG.root)},
```

```
}
display(pd.DataFrame([configuration_report]).T.rename(columns={0: "value"}))
```

```
                                                      value
workflow    {'model_version': 'resnet18-imagelevel-public-…
dataset     {'root_env_var': 'CASTING_DATA_ROOT', 'kaggle_…
training    {'image_size': 224, 'batch_size': 32, 'num_wor…
preprocess  {'use_clahe': True, 'denoise': False, 'clahe_c…
edge        {'p95_latency_budget_ms': 50.0, 'benchmark_rep…
artifact    {'root': 'artifacts/casting_inspection', 'save…
```

[3]:

```
# Code Intent:
# Create a compact run manifest that can travel with metrics and model␣
↪artifacts.
# Key Ideas:
# - Capture git commit when available without requiring the notebook to live␣
↪inside a repository.
# - Convert Paths, NumPy scalars, tuples, and nested config values into␣
↪JSON-safe objects.
# - Persist seed, device, package versions, and configuration in one object for␣
↪traceability.
# - Use the manifest later when saving artifacts so model, threshold, and␣
↪metrics remain linked.

# Run manifest utilities

def current_git_commit() -> Optional[str]:
    try:
        return subprocess.check_output(
            ["git", "rev-parse", "HEAD"],
            stderr=subprocess.DEVNULL,
            text=True,
        ).strip()
    except Exception:
        return None

def json_safe(payload):
    if isinstance(payload, Path):
        return str(payload)
    if isinstance(payload, dict):
        return {str(key): json_safe(value) for key, value in payload.items()}
    if isinstance(payload, (list, tuple)):
        return [json_safe(value) for value in payload]
    if isinstance(payload, np.generic):
        return payload.item()
    return payload
```

```
RUN_MANIFEST = {
    "workflow": "casting_inspection_industrial_workflow",
    "created_utc": datetime.now(timezone.utc).isoformat(),
    "seed": SEED,
    "device": str(DEVICE),
    "git_commit": current_git_commit(),
    "model_version": MODEL_VERSION,
    "preprocess_version": PREPROCESS_VERSION,
    "notebook_release_status": NOTEBOOK_RELEASE_STATUS,
    "leakage_policy": json_safe(LEAKAGE_POLICY),
    "package_versions": PACKAGE_VERSIONS,
    "configuration": json_safe(configuration_report),
}

display(pd.DataFrame([RUN_MANIFEST]).drop(columns=["package_versions",␣
↪"configuration"]))
```

```
                                workflow                      created_utc  \
0  casting_inspection_industrial_workflow  2026-05-01T02:21:16.467442+00:00

   seed device git_commit                       model_version  \
0    42   cuda       None  resnet18-imagelevel-public-pilot-v1

  preprocess_version                 notebook_release_status  \
0    opencv-clahe-v2  pilot_reference_not_production_approved

                                      leakage_policy
0  {'priority_order': ['train', 'val', 'test'], '…
```

1.3 3. Dataset Exploration

Dataset exploration is a production control point. The notebook validates folder layout, class balance, expected public counts, image readability, dimensions, exact hash leakage, and near-duplicate average-hash risk.

1.3.1 Colab Dataset Setup Behavior

This notebook now handles the most common Colab dataset failure mode. It first honors `CASTING_DATA_ROOT`, then searches common Kaggle, Colab, and KaggleHub cache locations. If the dataset is still missing and `AUTO_DOWNLOAD_CASTING_DATASET=True`, it attempts to download the public Kaggle dataset with `kagglehub`.

If an organization disables automatic downloads, set `AUTO_DOWNLOAD_CASTING_DATASET=False` and set `CASTING_DATA_ROOT` manually to the folder that contains `train/` and `test/`.

[4]:
```
# Code Intent:
# Resolve the physical dataset location in a way that works locally, on Kaggle,␣
↪and in Google Colab.
# Key Ideas:
```

```python
# - Prefer CASTING_DATA_ROOT when supplied because production pilots should use␣
↪explicit data roots.
# - Probe common Kaggle, Colab, local, and KaggleHub cache paths before␣
↪attempting downloads.
# - Recursively discover nested casting_data folders because KaggleHub often␣
↪caches datasets under versioned directories.
# - Optionally auto-download the public Kaggle dataset with kagglehub so Colab␣
↪runs do not fail at setup.
# - Display diagnostics for every candidate path and fail with actionable␣
↪guidance if download/search cannot resolve the data.

# Dataset root resolution

IMAGE_EXTENSIONS = set(DATASET_CONFIG.image_extensions)

def running_in_colab() -> bool:
    return importlib.util.find_spec("google.colab") is not None or Path("/
↪content").exists()

def is_dataset_root(path: Path) -> bool:
    return all((path / split / class_name).exists() for split in ["train",␣
↪"test"] for class_name in CLASS_TO_INDEX)

def unique_existing_paths(paths: Sequence[Path]) -> List[Path]:
    seen = set()
    unique = []
    for path in paths:
        try:
            resolved = path.expanduser().resolve()
        except Exception:
            resolved = path.expanduser()
        key = str(resolved)
        if key not in seen:
            seen.add(key)
            unique.append(resolved)
    return unique

def candidate_dataset_roots(extra_roots: Optional[Sequence[Path]] = None) ->␣
↪List[Path]:
    env_root = os.getenv(DATASET_CONFIG.root_env_var)
    candidates = []
    if env_root:
        env_path = Path(env_root)
        candidates.extend([env_path, env_path / "casting_data", env_path /␣
↪"casting_data" / "casting_data"])
```

```
    candidates.extend(
        [
            Path("/kaggle/input/real-life-industrial-dataset-of-casting-product/
↪casting_data/casting_data"),
            Path("/kaggle/input/real-life-industrial-dataset-of-casting-product/
↪casting_data"),
            Path("/content/casting_data/casting_data"),
            Path("/content/casting_data"),
            Path.cwd() / "casting_data" / "casting_data",
            Path.cwd() / "casting_data",
            Path.cwd(),
            Path.home() / ".cache" / "kagglehub" / "datasets" / "ravirajsinh45"␣
↪/ "real-life-industrial-dataset-of-casting-product",
        ]
    )

    if extra_roots:
        for root in extra_roots:
            candidates.extend([root, root / "casting_data", root /␣
↪"casting_data" / "casting_data"])

    return unique_existing_paths(candidates)

def discover_dataset_roots(search_bases: Sequence[Path], max_matches: int = 25)␣
↪-> List[Path]:
    discovered = []
    for base in unique_existing_paths(search_bases):
        if not base.exists() or not base.is_dir():
            continue
        direct_candidates = [base, base / "casting_data", base / "casting_data"␣
↪/ "casting_data"]
        for candidate in direct_candidates:
            if is_dataset_root(candidate):
                discovered.append(candidate)
        try:
            for casting_dir in base.rglob("casting_data"):
                for candidate in [casting_dir, casting_dir / "casting_data"]:
                    if is_dataset_root(candidate):
                        discovered.append(candidate)
                        if len(discovered) >= max_matches:
                            return unique_existing_paths(discovered)
        except (OSError, PermissionError):
            continue
    return unique_existing_paths(discovered)

def ensure_kagglehub_available() -> bool:
```

```
    if importlib.util.find_spec("kagglehub") is not None:
        return True
    if not DATASET_CONFIG.install_kagglehub_if_missing:
        return False
    print("kagglehub is not installed. Installing kagglehub for Colab dataset␣
↪download...")
    subprocess.check_call([sys.executable, "-m", "pip", "install", "-q",␣
↪"kagglehub"])
    importlib.invalidate_caches()
    return importlib.util.find_spec("kagglehub") is not None

def download_dataset_with_kagglehub() -> Optional[Path]:
    if not DATASET_CONFIG.auto_download_dataset:
        return None
    try:
        if not ensure_kagglehub_available():
            warnings.warn("kagglehub is unavailable. Automatic dataset download␣
↪was skipped.")
            return None
        import kagglehub

        print(f"Downloading Kaggle dataset: {DATASET_CONFIG.
↪kaggle_dataset_slug}")
        downloaded_path = Path(kagglehub.dataset_download(DATASET_CONFIG.
↪kaggle_dataset_slug)).resolve()
        print(f"KaggleHub download/cache path: {downloaded_path}")
        return downloaded_path
    except Exception as exc:
        warnings.warn(f"Automatic KaggleHub dataset download failed: {exc}")
        return None

def resolve_dataset_root() -> Path:
    diagnostics = []

    initial_candidates = candidate_dataset_roots()
    search_bases = initial_candidates + [Path("/content"), Path("/kaggle/
↪input"), Path.home() / ".cache" / "kagglehub"]
    discovered = discover_dataset_roots(search_bases)

    for path in unique_existing_paths(initial_candidates + discovered):
        diagnostics.append({"phase": "search", "candidate": str(path), "exists":
↪ path.exists(), "is_dataset_root": is_dataset_root(path)})
        if is_dataset_root(path):
            display(pd.DataFrame(diagnostics))
            return path.resolve()
```

```
    downloaded_path = download_dataset_with_kagglehub()
    if downloaded_path is not None:
        post_download_candidates =␣
↪candidate_dataset_roots(extra_roots=[downloaded_path])
        post_download_discovered = discover_dataset_roots([downloaded_path,␣
↪downloaded_path.parent, Path.home() / ".cache" / "kagglehub"])
        for path in unique_existing_paths(post_download_candidates +␣
↪post_download_discovered):
            diagnostics.append({"phase": "post_download", "candidate":␣
↪str(path), "exists": path.exists(), "is_dataset_root":␣
↪is_dataset_root(path)})
            if is_dataset_root(path):
                display(pd.DataFrame(diagnostics))
                return path.resolve()

    display(pd.DataFrame(diagnostics))
    searched = "\n".join(str(path) for path in␣
↪unique_existing_paths(candidate_dataset_roots()))
    raise FileNotFoundError(
        f"Could not find the casting dataset root. Set {DATASET_CONFIG.
↪root_env_var} to the directory containing "
        "train/ and test/ folders with def_front/ and ok_front/ subfolders. In␣
↪Colab, either allow the "
        "automatic KaggleHub download or run: \n"
        "    import os, kagglehub\n"
        f"    path = kagglehub.dataset_download('{DATASET_CONFIG.
↪kaggle_dataset_slug}')\n"
        f"    os.environ['{DATASET_CONFIG.root_env_var}'] = path\n\n"
        "Searched:\n" + searched
    )

DATA_ROOT = resolve_dataset_root()
RUN_MANIFEST["dataset_root"] = str(DATA_ROOT)
print(DATA_ROOT)
```

```
    phase                                          candidate  exists  \
0  search  /kaggle/input/real-life-industrial-dataset-of-…    True

   is_dataset_root
0             True
```

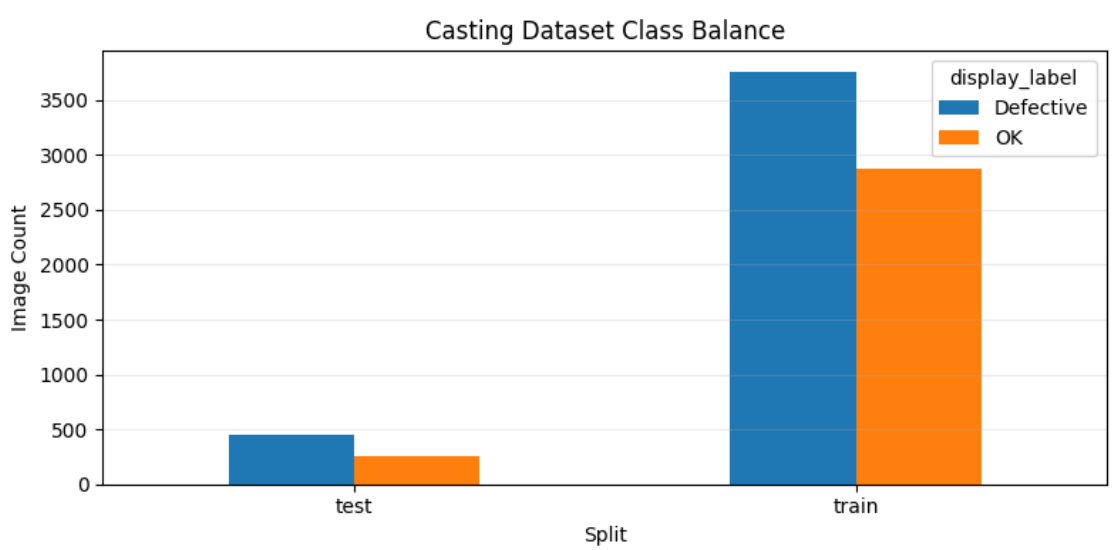
```
/kaggle/input/real-life-industrial-dataset-of-casting-
product/casting_data/casting_data
```

[5]:
```
# Code Intent:
# Convert the folder tree into a tabular image inventory and inspect class␣
↪balance.
```

```python
# Key Ideas:
# - Use the directory name as the source of truth for binary labels.
# - Keep file path, filename, split, class name, numeric label, and display␣
↪label together for auditability.
# - Warn when counts differ from the common public split without blocking␣
↪legitimate dataset variants.
# - Plot class balance because defect/OK imbalance affects sampling, loss␣
↪design, and threshold policy.

# Image inventory, expected counts, and class balance

def list_images(root: Path, split: str) -> pd.DataFrame:
    records = []
    for class_name, label in CLASS_TO_INDEX.items():
        class_dir = root / split / class_name
        for path in sorted(class_dir.rglob("*")):
            if path.suffix.lower() in IMAGE_EXTENSIONS:
                records.append(
                    {
                        "path": str(path),
                        "file_name": path.name,
                        "split": split,
                        "class_name": class_name,
                        "label": label,
                        "display_label": DISPLAY_NAMES[label],
                    }
                )
    return pd.DataFrame.from_records(records)

raw_train_df = list_images(DATA_ROOT, "train")
test_df = list_images(DATA_ROOT, "test")
dataset_df = pd.concat([raw_train_df, test_df], ignore_index=True)

if dataset_df.empty:
    raise ValueError(f"No image files found under {DATA_ROOT}")

class_counts = (
    dataset_df.groupby(["split", "class_name", "display_label"])
    .size()
    .reset_index(name="count")
    .sort_values(["split", "class_name"])
)
display(class_counts)

if DATASET_CONFIG.expected_public_counts_enabled:
    observed = {(row.split, row.class_name): int(row.count) for row in␣
↪class_counts.itertuples(index=False)}
```

```
    expected_warnings = []
    for key, expected_count in EXPECTED_PUBLIC_COUNTS.items():
        actual_count = observed.get(key)
        if actual_count != expected_count:
            expected_warnings.append({"split": key[0], "class_name": key[1],␣
 ↪"expected": expected_count, "actual": actual_count})
    if expected_warnings:
        warnings.warn("Dataset counts differ from the commonly documented␣
 ↪Kaggle split. Review before comparing metrics.")
        display(pd.DataFrame(expected_warnings))

pivot_counts = class_counts.pivot(index="split", columns="display_label",␣
 ↪values="count").fillna(0)
ax = pivot_counts.plot(kind="bar", figsize=(8, 4), rot=0)
ax.set_title("Casting Dataset Class Balance")
ax.set_xlabel("Split")
ax.set_ylabel("Image Count")
ax.grid(axis="y", alpha=0.25)
plt.tight_layout()
plt.show()
```

```
   split class_name display_label  count
0   test  def_front     Defective    453
1   test   ok_front            OK    262
2  train  def_front     Defective   3758
3  train   ok_front            OK   2875
```



[6]:
```
# Code Intent:
# Create a validation split and keep initial split accounting separate from␣
↪final leakage-filtered accounting.
# Key Ideas:
# - Use stratification so defective and OK proportions remain stable across␣
↪training and validation.
# - Keep the public test split isolated for final evaluation after threshold␣
↪selection.
# - Store initial split counts before leakage policy, then recompute final␣
↪counts after filtering.
# - Provide reusable split-summary helpers so downstream reporting always uses␣
↪the same contract.

# Train/validation split from provided training data

def build_model_split_frame(train_frame: pd.DataFrame, val_frame: pd.DataFrame,␣
↪test_frame: pd.DataFrame) -> pd.DataFrame:
    frames = []
    for split_name, frame in [("train", train_frame), ("val", val_frame),␣
↪("test", test_frame)]:
        working = frame.reset_index(drop=True).copy()
        working["model_split"] = split_name
        frames.append(working)
    return pd.concat(frames, ignore_index=True)

def build_split_count_table(train_frame: pd.DataFrame, val_frame: pd.DataFrame,␣
↪test_frame: pd.DataFrame) -> pd.DataFrame:
    split_summary = build_model_split_frame(train_frame, val_frame, test_frame)
    return (
        split_summary.groupby(["model_split", "display_label"])
        .size()
        .reset_index(name="count")
        .sort_values(["model_split", "display_label"])
    )

train_df, val_df = train_test_split(
    raw_train_df,
    test_size=DATASET_CONFIG.val_fraction_from_train,
    stratify=raw_train_df["label"],
    random_state=SEED,
)
train_df = train_df.reset_index(drop=True)
val_df = val_df.reset_index(drop=True)
test_df = test_df.reset_index(drop=True)

initial_split_count_table = build_split_count_table(train_df, val_df, test_df)
```

```
display(initial_split_count_table)
RUN_MANIFEST["initial_split_counts"] = initial_split_count_table.
↪to_dict(orient="records")
```

```
  model_split display_label  count
0        test     Defective    453
1        test            OK    262
2       train     Defective   3194
3       train            OK   2444
4         val     Defective    564
5         val            OK    431
```

[7]:
```
# Code Intent:
# Detect common industrial dataset quality risks before training.
# Key Ideas:
# - Audit image readability, shape, channels, and dtype to catch acquisition or␣
↪unpacking issues.
# - Compute SHA-1 hashes to identify exact duplicate files that cross train/
↪test boundaries.
# - Compute simple average hashes to flag possible near-duplicates across␣
↪splits.
# - Treat near-duplicate results as review evidence, not automatic deletion,␣
↪because augmented industrial images may be legitimate.
# - Add pre-policy audit results to the manifest, then final leakage status is␣
↪recomputed after filtering.

# Image readability, exact hash leakage, and near-duplicate average-hash checks

def inspect_image(path: str) -> Dict[str, object]:
    image = cv2.imread(path, cv2.IMREAD_UNCHANGED)
    if image is None:
        return {"readable": False, "height": None, "width": None, "channels":␣
↪None, "dtype": None}
    if image.ndim == 2:
        height, width = image.shape
        channels = 1
    else:
        height, width, channels = image.shape
    return {"readable": True, "height": int(height), "width": int(width),␣
↪"channels": int(channels), "dtype": str(image.dtype)}

def sha1_file(path: str, block_size: int = 1024 * 1024) -> Optional[str]:
    digest = hashlib.sha1()
    try:
        with open(path, "rb") as handle:
            while True:
                block = handle.read(block_size)
```

```
                if not block:
                    break
                digest.update(block)
        return digest.hexdigest()
    except OSError:
        return None

def average_hash(path: str, hash_size: int = 8) -> Optional[str]:
    image = cv2.imread(path, cv2.IMREAD_GRAYSCALE)
    if image is None:
        return None
    small = cv2.resize(image, (hash_size, hash_size), interpolation=cv2.
 ↪INTER_AREA)
    bits = (small > small.mean()).astype(np.uint8).flatten()
    value = 0
    for bit in bits:
        value = (value << 1) | int(bit)
    return f"{value:016x}"

def hamming_hex(left: str, right: str) -> int:
    return (int(left, 16) ^ int(right, 16)).bit_count()

def attach_image_hashes(df: pd.DataFrame, include_average_hash: bool = False)␣
 ↪-> pd.DataFrame:
    """Attach deterministic image hashes used for leakage auditing."""
    frame = df.reset_index(drop=True).copy()
    if "sha1" not in frame.columns or frame["sha1"].isna().any():
        frame["sha1"] = [sha1_file(path) for path in frame["path"]]
    if include_average_hash and ("ahash" not in frame.columns or frame["ahash"].
 ↪isna().any()):
        frame["ahash"] = [average_hash(path) for path in frame["path"]]
    return frame

def cross_split_overlap_frame(named_frames: Sequence[Tuple[str, pd.DataFrame]],␣
 ↪key_column: str) -> pd.DataFrame:
    rows = []
    prepared = []
    for split_name, frame in named_frames:
        if key_column not in frame.columns:
            frame = attach_image_hashes(frame) if key_column == "sha1" else␣
 ↪frame.copy()
        prepared.append((split_name, frame))
    for i, (left_name, left_frame) in enumerate(prepared):
        left_values = set(left_frame[key_column].dropna().tolist())
        for right_name, right_frame in prepared[i + 1:]:
            right_values = set(right_frame[key_column].dropna().tolist())
            overlap = sorted(left_values.intersection(right_values))
```

```
            if overlap:
                rows.append(
                    {
                        "left_split": left_name,
                        "right_split": right_name,
                        "key_column": key_column,
                        "overlap_count": len(overlap),
                        "examples": overlap[:10],
                    }
                )
    return pd.DataFrame(rows)

scan_df = dataset_df.copy()
if DATASET_CONFIG.max_audit_images is not None:
    scan_df = scan_df.sample(min(DATASET_CONFIG.max_audit_images,␣
 ↪len(scan_df)), random_state=SEED)

audit_df = pd.DataFrame([{"path": path, **inspect_image(path)} for path in␣
 ↪scan_df["path"].tolist()])
audit_summary = {
    "scanned_images": len(audit_df),
    "unreadable_images": int((~audit_df["readable"]).sum()),
    "unique_shape_count": int(audit_df[["height", "width", "channels"]].
 ↪drop_duplicates().shape[0]),
}
print(audit_summary)
display(audit_df[["height", "width", "channels", "dtype"]].value_counts().
 ↪reset_index(name="count").head(20))

dataset_df = attach_image_hashes(dataset_df)
exact_hash_groups = (
    dataset_df.dropna(subset=["sha1"])
    .groupby("sha1")
    .agg(split_count=("split", "nunique"), sample_count=("path", "count"),␣
 ↪examples=("path", lambda values: list(values)[:3]))
    .reset_index()
)
exact_hash_leaks = exact_hash_groups[(exact_hash_groups["split_count"] > 1) &␣
 ↪(exact_hash_groups["sample_count"] > 1)]
print(f"Pre-policy exact public train/test hash leak groups:␣
 ↪{len(exact_hash_leaks)}")
display(exact_hash_leaks.head(20))

hash_sample_df = dataset_df.sample(min(DATASET_CONFIG.
 ↪max_near_duplicate_images, len(dataset_df)), random_state=SEED).copy()
hash_sample_df = attach_image_hashes(hash_sample_df, include_average_hash=True)
```

```
hash_records = hash_sample_df.dropna(subset=["ahash"])[["path", "split",␣
↪"display_label", "ahash"]].to_dict("records")
near_duplicate_pairs = []
for i, left in enumerate(hash_records):
    for right in hash_records[i + 1:]:
        if left["split"] == right["split"]:
            continue
        distance = hamming_hex(left["ahash"], right["ahash"])
        if distance <= DATASET_CONFIG.near_duplicate_hamming_threshold:
            near_duplicate_pairs.append(
                {
                    "left_split": left["split"],
                    "right_split": right["split"],
                    "left_label": left["display_label"],
                    "right_label": right["display_label"],
                    "hamming": distance,
                    "left_path": left["path"],
                    "right_path": right["path"],
                }
            )
        if len(near_duplicate_pairs) >= 25:
            break
    if len(near_duplicate_pairs) >= 25:
        break

near_duplicate_df = pd.DataFrame(near_duplicate_pairs)
print(f"Near-duplicate cross-split average-hash pairs shown for review:␣
↪{len(near_duplicate_df)}")
display(near_duplicate_df)
RUN_MANIFEST["audit_summary"] = audit_summary
RUN_MANIFEST["pre_policy_exact_hash_leak_groups"] = int(len(exact_hash_leaks))
RUN_MANIFEST["near_duplicate_pairs_shown"] = int(len(near_duplicate_df))
```

```
{'scanned_images': 7348, 'unreadable_images': 0, 'unique_shape_count': 1}
```

```
   height  width  channels  dtype  count
0     300    300         3  uint8   7348
```

```
Pre-policy exact public train/test hash leak groups: 64
```

```
                                       sha1  split_count  sample_count  \
154   053c2bdec20390796a8454ae83b647c14ade04c            2             2
188   065593bfd02b435290141d4be3b8a034ec2e93a6            2             2
251   08c79bef293e30458e58c290e6ff26d67fc28b8a            2             2
274   09b1030d661d31f214919a3c6880189335ee56f2            2             2
305   0a7e040113f70e3ea05405c178ad746e9d9a532b            2             2
417   0e91985482f497c4ade0452631904a41e201b60c            2             2
537   12dadbb59bfcdd456e77e6acaabdd26f60b3f171            2             2
712   1903b34861b9d8e9568762e0952dc70b0e8ac648            2             2
```

```
718   1946bc9c44a82351df367583d9983db662b0c804            2             2
1109  272291cbd914ff084cd0a00adbe45962617582ee            2             2
1148  28ae60cc7feb25cc452abe5d73dc6cc8b77452bf            2             2
1408  319b1f5e7ca55d4ba99de2ddd73c9d3997898cb3            2             2
1424  31ff6cdd7be4cebe386fb91a6cdf122810b55aa0            2             2
1516  35699fd6a83acac14cf127d8c0d3a7b94f39e5e6            2             2
1518  3578d87a0f07a870a54065291053f209a90a63db            2             2
1658  3a05fe0550717183c48c2b12e048ad8f3c40239e            2             2
1770  3e56e52d0955c3345d8fe06073beaf0cd9c59b2c            2             2
1845  407bdb7eaf56636f7a686ef627a4b6f797fc6365            2             2
1846  4092d21d68409d1bc2765fb099a00e9050a88e7a            2             2
1916  42ed69e7631aee7a9ba007840aa52fba6c1a889a            2             2

                                                  examples
154   [/kaggle/input/real-life-industrial-dataset-of…
188   [/kaggle/input/real-life-industrial-dataset-of…
251   [/kaggle/input/real-life-industrial-dataset-of…
274   [/kaggle/input/real-life-industrial-dataset-of…
305   [/kaggle/input/real-life-industrial-dataset-of…
417   [/kaggle/input/real-life-industrial-dataset-of…
537   [/kaggle/input/real-life-industrial-dataset-of…
712   [/kaggle/input/real-life-industrial-dataset-of…
718   [/kaggle/input/real-life-industrial-dataset-of…
1109  [/kaggle/input/real-life-industrial-dataset-of…
1148  [/kaggle/input/real-life-industrial-dataset-of…
1408  [/kaggle/input/real-life-industrial-dataset-of…
1424  [/kaggle/input/real-life-industrial-dataset-of…
1516  [/kaggle/input/real-life-industrial-dataset-of…
1518  [/kaggle/input/real-life-industrial-dataset-of…
1658  [/kaggle/input/real-life-industrial-dataset-of…
1770  [/kaggle/input/real-life-industrial-dataset-of…
1845  [/kaggle/input/real-life-industrial-dataset-of…
1846  [/kaggle/input/real-life-industrial-dataset-of…
1916  [/kaggle/input/real-life-industrial-dataset-of…

Near-duplicate cross-split average-hash pairs shown for review: 25

   left_split right_split left_label right_label  hamming  \
0       train        test  Defective   Defective        4
1       train        test  Defective   Defective        2
2       train        test  Defective          OK        4
3       train        test  Defective          OK        3
4       train        test  Defective   Defective        4
5       train        test  Defective   Defective        4
6       train        test  Defective   Defective        4
7       train        test  Defective   Defective        3
8       train        test  Defective   Defective        3
9       train        test  Defective   Defective        3
10      train        test         OK   Defective        4
```

11	train	test	Defective	Defective	3
12	train	test	Defective	Defective	4
13	train	test	OK	Defective	3
14	train	test	OK	OK	3
15	train	test	OK	OK	4
16	train	test	Defective	OK	4
17	train	test	Defective	Defective	2
18	train	test	Defective	Defective	3
19	train	test	Defective	Defective	3
20	train	test	Defective	OK	3
21	train	test	Defective	OK	4
22	train	test	Defective	Defective	2
23	train	test	Defective	Defective	3
24	train	test	OK	Defective	3

	left_path \
0	/kaggle/input/real-life-industrial-dataset-of-…
1	/kaggle/input/real-life-industrial-dataset-of-…
2	/kaggle/input/real-life-industrial-dataset-of-…
3	/kaggle/input/real-life-industrial-dataset-of-…
4	/kaggle/input/real-life-industrial-dataset-of-…
5	/kaggle/input/real-life-industrial-dataset-of-…
6	/kaggle/input/real-life-industrial-dataset-of-…
7	/kaggle/input/real-life-industrial-dataset-of-…
8	/kaggle/input/real-life-industrial-dataset-of-…
9	/kaggle/input/real-life-industrial-dataset-of-…
10	/kaggle/input/real-life-industrial-dataset-of-…
11	/kaggle/input/real-life-industrial-dataset-of-…
12	/kaggle/input/real-life-industrial-dataset-of-…
13	/kaggle/input/real-life-industrial-dataset-of-…
14	/kaggle/input/real-life-industrial-dataset-of-…
15	/kaggle/input/real-life-industrial-dataset-of-…
16	/kaggle/input/real-life-industrial-dataset-of-…
17	/kaggle/input/real-life-industrial-dataset-of-…
18	/kaggle/input/real-life-industrial-dataset-of-…
19	/kaggle/input/real-life-industrial-dataset-of-…
20	/kaggle/input/real-life-industrial-dataset-of-…
21	/kaggle/input/real-life-industrial-dataset-of-…
22	/kaggle/input/real-life-industrial-dataset-of-…
23	/kaggle/input/real-life-industrial-dataset-of-…
24	/kaggle/input/real-life-industrial-dataset-of-…

	right_path
0	/kaggle/input/real-life-industrial-dataset-of-…
1	/kaggle/input/real-life-industrial-dataset-of-…
2	/kaggle/input/real-life-industrial-dataset-of-…
3	/kaggle/input/real-life-industrial-dataset-of-…
4	/kaggle/input/real-life-industrial-dataset-of-…

```
5   /kaggle/input/real-life-industrial-dataset-of-…
6   /kaggle/input/real-life-industrial-dataset-of-…
7   /kaggle/input/real-life-industrial-dataset-of-…
8   /kaggle/input/real-life-industrial-dataset-of-…
9   /kaggle/input/real-life-industrial-dataset-of-…
10  /kaggle/input/real-life-industrial-dataset-of-…
11  /kaggle/input/real-life-industrial-dataset-of-…
12  /kaggle/input/real-life-industrial-dataset-of-…
13  /kaggle/input/real-life-industrial-dataset-of-…
14  /kaggle/input/real-life-industrial-dataset-of-…
15  /kaggle/input/real-life-industrial-dataset-of-…
16  /kaggle/input/real-life-industrial-dataset-of-…
17  /kaggle/input/real-life-industrial-dataset-of-…
18  /kaggle/input/real-life-industrial-dataset-of-…
19  /kaggle/input/real-life-industrial-dataset-of-…
20  /kaggle/input/real-life-industrial-dataset-of-…
21  /kaggle/input/real-life-industrial-dataset-of-…
22  /kaggle/input/real-life-industrial-dataset-of-…
23  /kaggle/input/real-life-industrial-dataset-of-…
24  /kaggle/input/real-life-industrial-dataset-of-…
```

[8]:

```python
# Code Intent:
# Apply auditable split leakage policy before any model training occurs.
# Key Ideas:
# - Treat train as the anchor split, then remove downstream validation/test␣
↪rows that overlap by filename or exact SHA-1.
# - Recompute final split counts and final leakage checks after filtering.
# - Keep near-duplicate average-hash findings as warnings for quality review,␣
↪not automatic deletion.
# - Store final leakage policy, removals, split counts, and pass/fail status in␣
↪the run manifest.

# Final split leakage policy and assertions

def assert_no_filename_leakage(*named_frames: Tuple[str, pd.DataFrame]) -> None:
    leakage_df = cross_split_overlap_frame(named_frames, "file_name")
    if not leakage_df.empty:
        raise AssertionError(f"Potential filename leakage detected: {leakage_df.
↪to_dict(orient='records')}")

def assert_no_hash_leakage(*named_frames: Tuple[str, pd.DataFrame]) -> None:
    hashed_frames = [(name, attach_image_hashes(frame)) for name, frame in␣
↪named_frames]
    leakage_df = cross_split_overlap_frame(hashed_frames, "sha1")
    if not leakage_df.empty:
        raise AssertionError(f"Potential exact-hash leakage detected:␣
↪{leakage_df.to_dict(orient='records')}")
```

```python
def _drop_downstream_overlaps(
    candidate_frame: pd.DataFrame,
    candidate_name: str,
    upstream_frames: Sequence[Tuple[str, pd.DataFrame]],
) -> Tuple[pd.DataFrame, List[Dict[str, object]]]:
    upstream_filenames = set()
    upstream_hashes = set()
    upstream_names_by_hash = {}
    for upstream_name, upstream_frame in upstream_frames:
        upstream_filenames.update(upstream_frame["file_name"].dropna().tolist())
        for sha1 in upstream_frame["sha1"].dropna().tolist():
            upstream_hashes.add(sha1)
            upstream_names_by_hash.setdefault(sha1, set()).add(upstream_name)

    filename_overlap = candidate_frame["file_name"].isin(upstream_filenames)
    hash_overlap = candidate_frame["sha1"].notna() & candidate_frame["sha1"].
 ↪isin(upstream_hashes)
    remove_mask = filename_overlap | hash_overlap

    removal_records = []
    for _, row in candidate_frame.loc[remove_mask].iterrows():
        reasons = []
        if bool(row["file_name"] in upstream_filenames):
            reasons.append("filename_overlap")
        if pd.notna(row["sha1"]) and row["sha1"] in upstream_hashes:
            reasons.append("exact_hash_overlap")
        removal_records.append(
            {
                "removed_from": candidate_name,
                "file_name": row["file_name"],
                "display_label": row["display_label"],
                "sha1": row.get("sha1"),
                "reason": "+".join(reasons),
                "upstream_hash_splits": sorted(upstream_names_by_hash.get(row.
 ↪get("sha1"), [])),
                "path": row["path"],
            }
        )
    return candidate_frame.loc[~remove_mask].reset_index(drop=True),␣
 ↪removal_records

def apply_leakage_policy(
    train_frame: pd.DataFrame,
    val_frame: pd.DataFrame,
    test_frame: pd.DataFrame,
```

```
) -> Tuple[pd.DataFrame, pd.DataFrame, pd.DataFrame, pd.DataFrame, Dict[str,␣
↪object]]:
    train_hashed = attach_image_hashes(train_frame).reset_index(drop=True)
    val_hashed = attach_image_hashes(val_frame).reset_index(drop=True)
    test_hashed = attach_image_hashes(test_frame).reset_index(drop=True)

    filtered_val, val_removals = _drop_downstream_overlaps(
        val_hashed,
        "val",
        [("train", train_hashed)],
    )
    filtered_test, test_removals = _drop_downstream_overlaps(
        test_hashed,
        "test",
        [("train", train_hashed), ("val", filtered_val)],
    )
    removals_df = pd.DataFrame(val_removals + test_removals)

    final_named_frames = [("train", train_hashed), ("val", filtered_val),␣
↪("test", filtered_test)]
    final_filename_leaks = cross_split_overlap_frame(final_named_frames,␣
↪"file_name")
    final_hash_leaks = cross_split_overlap_frame(final_named_frames, "sha1")
    summary = {
        "policy": LEAKAGE_POLICY,
        "removed_from_validation": int(len(val_removals)),
        "removed_from_test": int(len(test_removals)),
        "final_filename_leak_pairs": int(final_filename_leaks["overlap_count"].
↪sum()) if not final_filename_leaks.empty else 0,
        "final_hash_leak_pairs": int(final_hash_leaks["overlap_count"].sum())␣
↪if not final_hash_leaks.empty else 0,
        "near_duplicate_action": LEAKAGE_POLICY["near_duplicate_overlap"],
    }
    return train_hashed, filtered_val, filtered_test, removals_df, summary

train_df, val_df, test_df, leakage_removals_df, leakage_summary =␣
↪apply_leakage_policy(train_df, val_df, test_df)
dataset_df = build_model_split_frame(train_df, val_df, test_df)
split_count_table = build_split_count_table(train_df, val_df, test_df)

if leakage_removals_df.empty:
    print("No downstream filename or exact-hash leakage removals were required.
↪")
else:
    print(
```

```
        f"Removed {len(leakage_removals_df)} downstream rows under leakage␣
↪policy: "
        f"{leakage_summary['removed_from_validation']} from validation,␣
↪{leakage_summary['removed_from_test']} from test."
    )
    display(leakage_removals_df.groupby(["removed_from", "reason",␣
↪"display_label"]).size().reset_index(name="count"))
    display(leakage_removals_df.head(20))

assert_no_filename_leakage(("train", train_df), ("val", val_df), ("test",␣
↪test_df))
assert_no_hash_leakage(("train", train_df), ("val", val_df), ("test", test_df))
print("Final filename and exact-hash leakage checks passed across train/val/
↪test.")
display(split_count_table)

RUN_MANIFEST["final_split_counts"] = split_count_table.to_dict(orient="records")
RUN_MANIFEST["split_counts"] = RUN_MANIFEST["final_split_counts"]
RUN_MANIFEST["leakage_policy"] = json_safe(LEAKAGE_POLICY)
RUN_MANIFEST["leakage_summary"] = json_safe(leakage_summary)
RUN_MANIFEST["leakage_removal_counts"] = leakage_removals_df.
↪groupby(["removed_from", "reason"]).size().reset_index(name="count").
↪to_dict(orient="records") if not leakage_removals_df.empty else []
```

```
Removed 144 downstream rows under leakage policy: 0 from validation, 144 from
test.

  removed_from                                reason display_label  count
0         test                      filename_overlap     Defective     44
1         test                      filename_overlap            OK     36
2         test  filename_overlap+exact_hash_overlap            OK     64

   removed_from            file_name display_label  \
0          test  cast_ok_0_1001.jpeg            OK
1          test  cast_ok_0_1002.jpeg            OK
2          test  cast_ok_0_1003.jpeg            OK
3          test  cast_ok_0_1019.jpeg            OK
4          test  cast_ok_0_1020.jpeg            OK
5          test  cast_ok_0_1021.jpeg            OK
6          test  cast_ok_0_1022.jpeg            OK
7          test  cast_ok_0_1026.jpeg            OK
8          test  cast_ok_0_1028.jpeg            OK
9          test  cast_ok_0_1031.jpeg            OK
10         test  cast_ok_0_1040.jpeg            OK
11         test  cast_ok_0_1066.jpeg            OK
12         test  cast_ok_0_1069.jpeg            OK
13         test  cast_ok_0_1077.jpeg            OK
14         test  cast_ok_0_1088.jpeg            OK
```

```
15         test  cast_ok_0_1092.jpeg            OK
16         test  cast_ok_0_1093.jpeg            OK
17         test  cast_ok_0_1095.jpeg            OK
18         test  cast_ok_0_1096.jpeg            OK
19         test  cast_ok_0_1120.jpeg            OK

                                          sha1  \
0   c5f1686ec14d615ad56d91c5e30f8a301e56503b
1   d23fd56c9981794307538f9453869861bcbfed02
2   f4497a1c9c27c04b67b888eefecfc6292b0bc48b
3   272291cbd914ff084cd0a00adbe45962617582ee
4   f4f692118ebf0cf584ab2ab9c8651ce02ee27ba7
5   3e56e52d0955c3345d8fe06073beaf0cd9c59b2c
6   5e37e592cbbb3f476527ed85e7035b3fcbc0de23
7   dd9941be9421c78b7a541fd17093ef979260c698
8   48f8aecc468ccc687afe3e852bd25d0ceab7ecee
9   6dbfb0a89ff7f6d38d3369dba3d548ec8f186a75
10  a213b6120ba323af89b406bd49551560e1224d61
11  c969bf96871db1df6b26864937ee01f0ea0d2893
12  adf01ab22cbb3895db305417ca5f9cea90d5cb51
13  6d7992a020851b6194e1fbc6c061186830127670
14  c14a5a45d164b2e833a86d3b511b529bba1ac11e
15  e104a3dbafec8c4a342b4e96b0595893fd4e906a
16  b89675bc28b8ae7a7e8d29ecb74bd6df7fc09cd1
17  31ff6cdd7be4cebe386fb91a6cdf122810b55aa0
18  a5a15a41b1e5ff16c1fafd659a41febc0e761440
19  706151e9e52fe7c231aeb23408d38ed13f4414d2

                                reason upstream_hash_splits  \
0   filename_overlap+exact_hash_overlap                [val]
1   filename_overlap+exact_hash_overlap              [train]
2   filename_overlap+exact_hash_overlap              [train]
3   filename_overlap+exact_hash_overlap              [train]
4   filename_overlap+exact_hash_overlap              [train]
5   filename_overlap+exact_hash_overlap              [train]
6   filename_overlap+exact_hash_overlap              [train]
7   filename_overlap+exact_hash_overlap              [train]
8   filename_overlap+exact_hash_overlap              [train]
9   filename_overlap+exact_hash_overlap                [val]
10  filename_overlap+exact_hash_overlap              [train]
11  filename_overlap+exact_hash_overlap              [train]
12  filename_overlap+exact_hash_overlap              [train]
13  filename_overlap+exact_hash_overlap              [train]
14  filename_overlap+exact_hash_overlap              [train]
15  filename_overlap+exact_hash_overlap              [train]
16  filename_overlap+exact_hash_overlap              [train]
17  filename_overlap+exact_hash_overlap              [train]
18  filename_overlap+exact_hash_overlap              [train]
```

```
19  filename_overlap+exact_hash_overlap              [train]

                                                 path
0   /kaggle/input/real-life-industrial-dataset-of-…
1   /kaggle/input/real-life-industrial-dataset-of-…
2   /kaggle/input/real-life-industrial-dataset-of-…
3   /kaggle/input/real-life-industrial-dataset-of-…
4   /kaggle/input/real-life-industrial-dataset-of-…
5   /kaggle/input/real-life-industrial-dataset-of-…
6   /kaggle/input/real-life-industrial-dataset-of-…
7   /kaggle/input/real-life-industrial-dataset-of-…
8   /kaggle/input/real-life-industrial-dataset-of-…
9   /kaggle/input/real-life-industrial-dataset-of-…
10  /kaggle/input/real-life-industrial-dataset-of-…
11  /kaggle/input/real-life-industrial-dataset-of-…
12  /kaggle/input/real-life-industrial-dataset-of-…
13  /kaggle/input/real-life-industrial-dataset-of-…
14  /kaggle/input/real-life-industrial-dataset-of-…
15  /kaggle/input/real-life-industrial-dataset-of-…
16  /kaggle/input/real-life-industrial-dataset-of-…
17  /kaggle/input/real-life-industrial-dataset-of-…
18  /kaggle/input/real-life-industrial-dataset-of-…
19  /kaggle/input/real-life-industrial-dataset-of-…

Final filename and exact-hash leakage checks passed across train/val/test.

  model_split display_label  count
0        test     Defective    409
1        test            OK    162
2       train     Defective   3194
3       train            OK   2444
4         val     Defective    564
5         val            OK    431
```

[9]:
```
# Code Intent:
# Show representative OK and defective images for visual sanity checking.
# Key Ideas:
# - Read images through OpenCV and convert BGR to RGB for correct display.
# - Sample equally by class to avoid visual review being dominated by the␣
 ↪larger class.
# - Use compact grids so senior reviewers can quickly assess lighting, fixture,␣
 ↪and defect cues.

# Sample visualization

def read_rgb(path: str) -> np.ndarray:
    image = cv2.imread(path, cv2.IMREAD_COLOR)
    if image is None:
```

```
        raise ValueError(f"Unreadable image: {path}")
    return cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

def plot_samples(df: pd.DataFrame, samples_per_class: int = 6, title: str =␣
 ↪"Sample Images") -> None:
    min_count = int(df.groupby("display_label").size().min())
    n_per_class = min(samples_per_class, min_count)
    selected = (
        df.groupby("display_label", group_keys=False)
        .sample(n=n_per_class, random_state=SEED)
        .reset_index(drop=True)
    )
    cols = n_per_class
    rows = int(np.ceil(len(selected) / max(cols, 1)))
    fig, axes = plt.subplots(rows, cols, figsize=(cols * 2.2, rows * 2.5))
    axes = np.array(axes).reshape(-1)
    fig.suptitle(title, fontsize=14)
    for ax, (_, row) in zip(axes, selected.iterrows()):
        ax.imshow(read_rgb(row["path"]), cmap="gray")
        ax.set_title(row["display_label"], fontsize=10)
        ax.axis("off")
    for ax in axes[len(selected):]:
        ax.axis("off")
    plt.tight_layout()
    plt.show()

plot_samples(raw_train_df, samples_per_class=6, title="Raw Training Samples by␣
 ↪Label")
```

1.3.2 Defect Characteristics Discussion

The public labels should be treated conservatively. A classifier can learn genuine defect evidence, but it can also learn camera, lighting, fixture, background, compression, or augmentation artifacts. Production review must inspect false accepts and false rejects, and separate process-relevant casting evidence from image-acquisition shortcuts.

1.4 4. Preprocessing & Augmentation

Preprocessing must match deployment. Training augmentation improves robustness, but deployment preprocessing must stay deterministic, inspectable, and inexpensive on edge hardware.

Baseline preprocessing: OpenCV load, grayscale conversion, optional CLAHE, fixed resize, and 3-channel RGB conversion for ImageNet-pretrained ResNet compatibility.

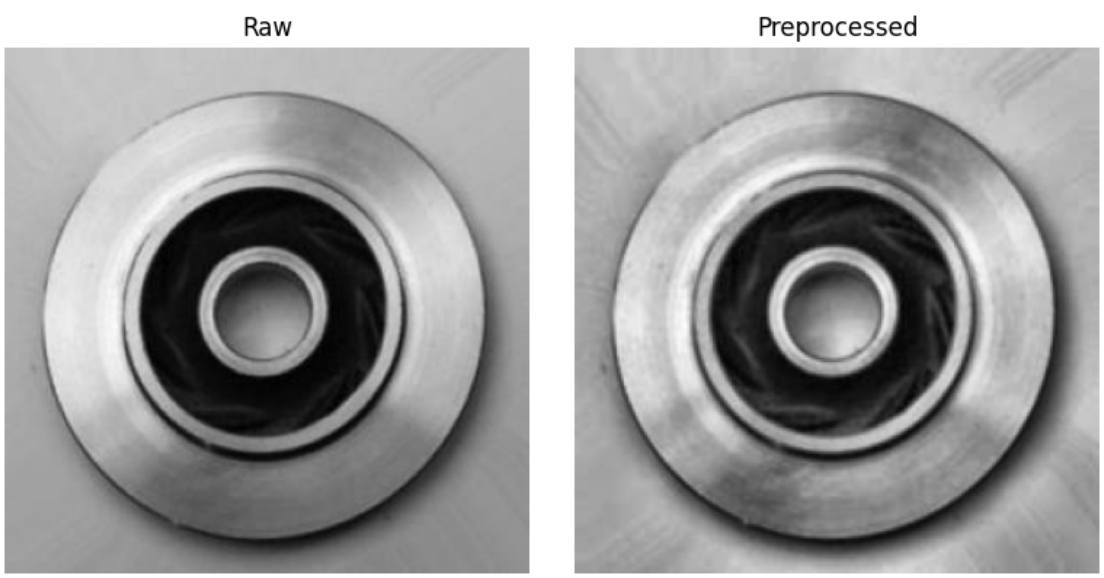
Training augmentation remains manufacturing-relevant: small rotation/translation, mild brightness/contrast changes, occasional blur, and orientation-safe horizontal flips. Aggressive crops are avoided because they can erase small defects.

[10]:
```
# Code Intent:
# Define deterministic preprocessing and lightweight diagnostics for edge␣
↪deployment.
# Key Ideas:
# - Convert to grayscale because the dataset is effectively grayscale␣
↪industrial imagery.
# - Apply CLAHE only through configuration so contrast normalization is␣
↪explicit and reproducible.
# - Resize deterministically to the model input size and return 3-channel RGB␣
↪for ImageNet ResNet weights.
# - Compute intensity, edge, and sharpness diagnostics for drift monitoring␣
↪rather than using them as model inputs.

# OpenCV preprocessing and image-quality diagnostics

def preprocess_for_inspection(
    image_bgr: np.ndarray,
    output_size: int = TRAINING_CONFIG.image_size,
    config: PreprocessConfig = PREPROCESS_CONFIG,
) -> np.ndarray:
    '''
    Deterministic edge-compatible preprocessing.
    Returns RGB uint8 image with shape (output_size, output_size, 3).
    '''
    if image_bgr is None:
        raise ValueError("Input image is None")

    gray = cv2.cvtColor(image_bgr, cv2.COLOR_BGR2GRAY) if image_bgr.ndim == 3␣
↪else image_bgr
```

```
    if config.denoise:
        gray = cv2.fastNlMeansDenoising(gray, h=5)

    if config.use_clahe:
        clahe = cv2.createCLAHE(clipLimit=config.clahe_clip_limit,␣
↪tileGridSize=config.clahe_tile_grid_size)
        gray = clahe.apply(gray)

    resized = cv2.resize(gray, (output_size, output_size), interpolation=cv2.
↪INTER_AREA)
    return cv2.cvtColor(resized, cv2.COLOR_GRAY2RGB)

def texture_diagnostics(image_bgr: np.ndarray) -> Dict[str, float]:
    '''
    Lightweight diagnostics for drift monitoring. These are not model features␣
↪by default.
    '''
    if image_bgr is None:
        raise ValueError("Input image is None")
    gray = cv2.cvtColor(image_bgr, cv2.COLOR_BGR2GRAY) if image_bgr.ndim == 3␣
↪else image_bgr
    edges = cv2.Canny(gray, threshold1=50, threshold2=150)
    laplacian_variance = cv2.Laplacian(gray, cv2.CV_64F).var()
    return {
        "mean_intensity": float(np.mean(gray)),
        "std_intensity": float(np.std(gray)),
        "edge_density": float(np.mean(edges > 0)),
        "laplacian_variance": float(laplacian_variance),
    }
```

[11]:
```
# Code Intent:
# Verify preprocessing visually and programmatically before model training.
# Key Ideas:
# - Compare raw and processed images to catch destructive contrast or resizing␣
↪choices.
# - Assert shape, dtype, value range, and finite values as a deployment␣
↪contract.
# - Sample multiple images so preprocessing failures are caught before training␣
↪begins.

# Preprocessing visual and contract checks

sample_path = raw_train_df.sample(1, random_state=SEED)["path"].iloc[0]
sample_bgr = cv2.imread(sample_path, cv2.IMREAD_COLOR)
sample_processed = preprocess_for_inspection(sample_bgr)
```

```
fig, axes = plt.subplots(1, 2, figsize=(8, 4))
axes[0].imshow(cv2.cvtColor(sample_bgr, cv2.COLOR_BGR2RGB))
axes[0].set_title("Raw")
axes[0].axis("off")
axes[1].imshow(sample_processed)
axes[1].set_title("Preprocessed")
axes[1].axis("off")
plt.tight_layout()
plt.show()

display(pd.DataFrame([texture_diagnostics(sample_bgr)]))

def validate_preprocessing(path: str) -> None:
    image = cv2.imread(path, cv2.IMREAD_COLOR)
    output = preprocess_for_inspection(image)
    assert output.shape == (TRAINING_CONFIG.image_size, TRAINING_CONFIG.
 ↪image_size, 3), output.shape
    assert output.dtype == np.uint8, output.dtype
    assert output.min() >= 0 and output.max() <= 255
    assert np.isfinite(output).all()

for path in raw_train_df.sample(min(50, len(raw_train_df)),␣
 ↪random_state=SEED)["path"]:
    validate_preprocessing(path)

print("Preprocessing contract validation passed.")
```




```
  mean_intensity  std_intensity  edge_density  laplacian_variance
```

```
0       142.365678       63.810556       0.041433           153.846708
```

```
Preprocessing contract validation passed.
```

[12]:
```
# Code Intent:
# Define train-time augmentation and eval-time normalization consistently.
# Key Ideas:
# - Use modest geometric and photometric augmentation aligned with␣
↪manufacturing variation.
# - Avoid aggressive crops because they can remove small defects and corrupt␣
↪labels.
# - Normalize with ImageNet statistics because the ResNet backbone uses␣
↪ImageNet-pretrained weights.
# - Keep evaluation transforms deterministic so validation/test metrics reflect␣
↪model behavior, not augmentation noise.

# TorchVision transforms

IMAGENET_MEAN = [0.485, 0.456, 0.406]
IMAGENET_STD = [0.229, 0.224, 0.225]

train_transform = transforms.Compose(
    [
        transforms.RandomAffine(degrees=7, translate=(0.02, 0.02), scale=(0.97,␣
↪1.03)),
        transforms.RandomApply([transforms.ColorJitter(brightness=0.15,␣
↪contrast=0.15)], p=0.5),
        transforms.RandomApply([transforms.GaussianBlur(kernel_size=3, sigma=(0.
↪1, 1.0))], p=0.15),
        transforms.RandomHorizontalFlip(p=0.5),
        transforms.ToTensor(),
        transforms.Normalize(mean=IMAGENET_MEAN, std=IMAGENET_STD),
    ]
)

eval_transform = transforms.Compose(
    [
        transforms.ToTensor(),
        transforms.Normalize(mean=IMAGENET_MEAN, std=IMAGENET_STD),
    ]
)
```

1.5 5. Baseline Classification Model

The baseline uses ResNet18 transfer learning. This is a conservative production-pilot choice: mature architecture, stable PyTorch/TorchVision implementation, fast batch-size-1 inference, straightforward export path, and direct compatibility with image-level labels.

The positive class is **defective**. Defect recall and false-accept count are primary governance metrics.

[13]:
```
# Code Intent:
# Wrap the image inventory in a PyTorch Dataset and build loaders for training,␣
↪validation, and test.
# Key Ideas:
# - Load every image through the same OpenCV preprocessing path intended for␣
↪deployment.
# - Convert the preprocessed image into PIL only because TorchVision transforms␣
↪expect that interface here.
# - Use a weighted sampler to reduce class imbalance pressure during training.
# - Keep validation and test loaders deterministic with shuffle disabled.

# Dataset and dataloaders

class CastingClassificationDataset(Dataset):
    def __init__(
        self,
        df: pd.DataFrame,
        transform=None,
        preprocess_config: PreprocessConfig = PREPROCESS_CONFIG,
    ):
        self.df = df.reset_index(drop=True).copy()
        self.transform = transform
        self.preprocess_config = preprocess_config

    def __len__(self) -> int:
        return len(self.df)

    def __getitem__(self, index: int):
        row = self.df.iloc[index]
        image_bgr = cv2.imread(row["path"], cv2.IMREAD_COLOR)
        if image_bgr is None:
            raise ValueError(f"Unreadable image: {row['path']}")
        image_rgb = preprocess_for_inspection(
            image_bgr,
            output_size=TRAINING_CONFIG.image_size,
            config=self.preprocess_config,
        )
        image_pil = Image.fromarray(image_rgb)
        image_tensor = self.transform(image_pil) if self.transform else␣
↪transforms.ToTensor()(image_pil)
        label = torch.tensor(int(row["label"]), dtype=torch.long)
        return image_tensor, label

def make_weighted_sampler(df: pd.DataFrame) -> WeightedRandomSampler:
    label_counts = df["label"].value_counts().to_dict()
```

```
    sample_weights = df["label"].map(lambda label: 1.0 /␣
 ↪label_counts[int(label)]).values
    return WeightedRandomSampler(
        weights=torch.DoubleTensor(sample_weights),
        num_samples=len(sample_weights),
        replacement=True,
    )

train_dataset = CastingClassificationDataset(train_df,␣
 ↪transform=train_transform)
val_dataset = CastingClassificationDataset(val_df, transform=eval_transform)
test_dataset = CastingClassificationDataset(test_df, transform=eval_transform)

train_sampler = make_weighted_sampler(train_df) if TRAINING_CONFIG.
 ↪use_weighted_sampler else None

train_loader = DataLoader(
    train_dataset,
    batch_size=TRAINING_CONFIG.batch_size,
    sampler=train_sampler,
    shuffle=train_sampler is None,
    num_workers=TRAINING_CONFIG.num_workers,
    pin_memory=torch.cuda.is_available(),
)
val_loader = DataLoader(
    val_dataset,
    batch_size=TRAINING_CONFIG.batch_size,
    shuffle=False,
    num_workers=TRAINING_CONFIG.num_workers,
    pin_memory=torch.cuda.is_available(),
)
test_loader = DataLoader(
    test_dataset,
    batch_size=TRAINING_CONFIG.batch_size,
    shuffle=False,
    num_workers=TRAINING_CONFIG.num_workers,
    pin_memory=torch.cuda.is_available(),
)

print(f"Train batches: {len(train_loader)}")
print(f"Val batches:   {len(val_loader)}")
print(f"Test batches:  {len(test_loader)}")
```

```
Train batches: 177
Val batches:   32
Test batches:  18
```

[14]:
```
# Code Intent:
# Define the ResNet classifier, optimization policy, and reusable metric␣
↪functions.
# Key Ideas:
# - Replace the ImageNet classification head with a two-class inspection head.
# - Keep optional backbone freezing available for low-data or fast-pilot␣
↪experiments.
# - Use AdamW with cosine scheduling as a stable transfer-learning baseline.
# - Report defect precision, recall, and F1 explicitly because accuracy alone␣
↪is insufficient for inspection.
# - Keep confusion-matrix formatting standardized for review and acceptance␣
↪reports.

# Model, loss, optimizer, and core metric utilities

def build_resnet18_classifier(num_classes: int = 2, freeze_backbone: bool =␣
↪False) -> nn.Module:
    weights = models.ResNet18_Weights.IMAGENET1K_V1
    model = models.resnet18(weights=weights)
    if freeze_backbone:
        for parameter in model.parameters():
            parameter.requires_grad = False
    in_features = model.fc.in_features
    model.fc = nn.Sequential(nn.Dropout(p=0.2), nn.Linear(in_features,␣
↪num_classes))
    return model

def make_loss(df: pd.DataFrame) -> nn.Module:
    if not TRAINING_CONFIG.use_class_weighted_loss:
        return nn.CrossEntropyLoss()
    label_counts = df["label"].value_counts().sort_index()
    weights = torch.tensor([1.0 / label_counts.get(i, 1) for i in range(2)],␣
↪dtype=torch.float32)
    weights = weights / weights.sum() * 2
    return nn.CrossEntropyLoss(weight=weights.to(DEVICE))

model = build_resnet18_classifier(num_classes=2, freeze_backbone=False).
↪to(DEVICE)
criterion = make_loss(train_df)
optimizer = torch.optim.AdamW(model.parameters(), lr=TRAINING_CONFIG.
↪learning_rate, weight_decay=TRAINING_CONFIG.weight_decay)
scheduler = torch.optim.lr_scheduler.CosineAnnealingLR(optimizer,␣
↪T_max=max(TRAINING_CONFIG.epochs, 1))

def compute_binary_metrics(y_true: np.ndarray, y_pred: np.ndarray) -> Dict[str,␣
↪float]:
```

```
    return {
        "accuracy": accuracy_score(y_true, y_pred),
        "precision_defect": precision_score(y_true, y_pred,␣
↪pos_label=TRAINING_CONFIG.defect_positive_label, zero_division=0),
        "recall_defect": recall_score(y_true, y_pred, pos_label=TRAINING_CONFIG.
↪defect_positive_label, zero_division=0),
        "f1_defect": f1_score(y_true, y_pred, pos_label=TRAINING_CONFIG.
↪defect_positive_label, zero_division=0),
    }

def confusion_matrix_df(y_true: np.ndarray, y_pred: np.ndarray) -> pd.DataFrame:
    matrix = confusion_matrix(y_true, y_pred, labels=[0, 1])
    return pd.DataFrame(matrix, index=["Actual OK", "Actual Defective"],␣
↪columns=["Predicted OK", "Predicted Defective"])

model.fc
```

```
Downloading: "https://download.pytorch.org/models/resnet18-f37072fd.pth" to
/root/.cache/torch/hub/checkpoints/resnet18-f37072fd.pth

100%|      | 44.7M/44.7M [00:00<00:00, 70.7MB/s]
```

[14]:
```
Sequential(
  (0): Dropout(p=0.2, inplace=False)
  (1): Linear(in_features=512, out_features=2, bias=True)
)
```

[15]:
```
# Code Intent:
# Train the classifier and retain the validation-best checkpoint.
# Key Ideas:
# - Train with standard supervised cross-entropy over image-level labels.
# - Select checkpoints using a defect-recall-heavy score because false accepts␣
↪are the key quality risk.
# - Preserve the best model state in memory before threshold governance and␣
↪final test evaluation.
# - Keep prediction output as true labels, argmax labels, and probabilities so␣
↪threshold analysis is possible.

# Training and prediction loops

def train_one_epoch(model: nn.Module, loader: DataLoader) -> Dict[str, float]:
    model.train()
    running_loss = 0.0
    total = 0
    for images, labels in loader:
        images = images.to(DEVICE, non_blocking=True)
        labels = labels.to(DEVICE, non_blocking=True)
        optimizer.zero_grad(set_to_none=True)
```

```
        logits = model(images)
        loss = criterion(logits, labels)
        loss.backward()
        optimizer.step()
        batch_size = labels.size(0)
        running_loss += loss.item() * batch_size
        total += batch_size
    return {"loss": running_loss / max(total, 1)}

@torch.no_grad()
def predict_loader(model: nn.Module, loader: DataLoader) -> Tuple[np.ndarray,␣
 ↪np.ndarray, np.ndarray]:
    model.eval()
    y_true, y_pred, y_prob = [], [], []
    for images, labels in loader:
        images = images.to(DEVICE, non_blocking=True)
        logits = model(images)
        probabilities = torch.softmax(logits, dim=1)
        predictions = torch.argmax(probabilities, dim=1)
        y_true.append(labels.cpu().numpy())
        y_pred.append(predictions.cpu().numpy())
        y_prob.append(probabilities.cpu().numpy())
    return np.concatenate(y_true), np.concatenate(y_pred), np.
 ↪concatenate(y_prob)

history = []
best_state = None
best_score = -np.inf

for epoch in range(1, TRAINING_CONFIG.epochs + 1):
    train_metrics = train_one_epoch(model, train_loader)
    scheduler.step()
    val_true_argmax, val_pred_argmax, val_prob_argmax = predict_loader(model,␣
 ↪val_loader)
    val_metrics_argmax = compute_binary_metrics(val_true_argmax,␣
 ↪val_pred_argmax)
    selection_score = val_metrics_argmax["recall_defect"] * 10.0 +␣
 ↪val_metrics_argmax["f1_defect"]
    if selection_score > best_score:
        best_score = selection_score
        best_state = copy.deepcopy(model.state_dict())
    row = {
        "epoch": epoch,
        "train_loss": train_metrics["loss"],
        "lr": scheduler.get_last_lr()[0],
        **{f"val_argmax_{key}": value for key, value in val_metrics_argmax.
 ↪items()},
```

```
    }
    history.append(row)
    print(row)

if best_state is not None:
    model.load_state_dict(best_state)

history_df = pd.DataFrame(history)
display(history_df)
```

```
{'epoch': 1, 'train_loss': 0.08701123334910112, 'lr': 9.045084971874738e-05,
'val_argmax_accuracy': 0.9959798994974874, 'val_argmax_precision_defect': 1.0,
'val_argmax_recall_defect': 0.9929078014184397, 'val_argmax_f1_defect':
0.99644128113879}
{'epoch': 2, 'train_loss': 0.020040267080718762, 'lr': 6.545084971874737e-05,
'val_argmax_accuracy': 0.9969849246231156, 'val_argmax_precision_defect': 1.0,
'val_argmax_recall_defect': 0.9946808510638298, 'val_argmax_f1_defect':
0.9973333333333333}
{'epoch': 3, 'train_loss': 0.016859391140419514, 'lr': 3.454915028125263e-05,
'val_argmax_accuracy': 0.9979899497487437, 'val_argmax_precision_defect': 1.0,
'val_argmax_recall_defect': 0.9964539007092199, 'val_argmax_f1_defect':
0.9982238010657194}
{'epoch': 4, 'train_loss': 0.00912075937625996, 'lr': 9.549150281252631e-06,
'val_argmax_accuracy': 0.9979899497487437, 'val_argmax_precision_defect': 1.0,
'val_argmax_recall_defect': 0.9964539007092199, 'val_argmax_f1_defect':
0.9982238010657194}
{'epoch': 5, 'train_loss': 0.008532325859375182, 'lr': 0.0,
'val_argmax_accuracy': 0.9979899497487437, 'val_argmax_precision_defect': 1.0,
'val_argmax_recall_defect': 0.9964539007092199, 'val_argmax_f1_defect':
0.9982238010657194}
```

```
   epoch  train_loss        lr  val_argmax_accuracy  \
0      1    0.087011  0.000090             0.995980
1      2    0.020040  0.000065             0.996985
2      3    0.016859  0.000035             0.997990
3      4    0.009121  0.000010             0.997990
4      5    0.008532  0.000000             0.997990

   val_argmax_precision_defect  val_argmax_recall_defect  val_argmax_f1_defect
0                          1.0                  0.992908              0.996441
1                          1.0                  0.994681              0.997333
2                          1.0                  0.996454              0.998224
3                          1.0                  0.996454              0.998224
4                          1.0                  0.996454              0.998224
```

[16]:
```
# Code Intent:
# Convert model probabilities into a governed inspection operating point.
# Key Ideas:
```

```
# - Sweep thresholds on validation data rather than accepting default argmax␣
↪behavior.
# - Select a conservative threshold that meets defect-recall policy while␣
↪minimizing defective parts accepted as OK.
# - Add ROC-AUC, PR-AUC, Brier score, confidence intervals, and␣
↪false-accept-rate framing.
# - Store threshold policy as data so the chosen operating point can be␣
↪versioned with artifacts.

# Advanced evaluation, calibration, threshold governance, and uncertainty bands

def wilson_interval(successes: int, total: int, z: float = 1.96) ->␣
↪Tuple[float, float]:
    if total <= 0:
        return (np.nan, np.nan)
    p_hat = successes / total
    denominator = 1.0 + z**2 / total
    center = (p_hat + z**2 / (2 * total)) / denominator
    half_width = z * math.sqrt((p_hat * (1.0 - p_hat) + z**2 / (4 * total)) /␣
↪total) / denominator
    return (float(max(0.0, center - half_width)), float(min(1.0, center +␣
↪half_width)))

def compute_advanced_metrics(y_true: np.ndarray, defect_prob: np.ndarray,␣
↪threshold: float) -> Dict[str, float]:
    y_pred = (defect_prob >= threshold).astype(int)
    metrics = compute_binary_metrics(y_true, y_pred)
    defect_mask = y_true == TRAINING_CONFIG.defect_positive_label
    ok_mask = y_true != TRAINING_CONFIG.defect_positive_label
    false_accepts = int((defect_mask & (y_pred == 0)).sum())
    false_rejects = int((ok_mask & (y_pred == 1)).sum())
    defect_support = int(defect_mask.sum())
    ok_support = int(ok_mask.sum())
    if len(np.unique(y_true)) > 1:
        metrics["roc_auc"] = roc_auc_score(y_true, defect_prob)
        metrics["pr_auc"] = average_precision_score(y_true, defect_prob)
    else:
        metrics["roc_auc"] = np.nan
        metrics["pr_auc"] = np.nan
    metrics["brier_score"] = brier_score_loss(y_true, defect_prob)
    metrics["false_accepts"] = false_accepts
    metrics["false_rejects"] = false_rejects
    metrics["defect_support"] = defect_support
    metrics["ok_support"] = ok_support
    metrics["false_accept_rate"] = false_accepts / max(defect_support, 1)
    metrics["false_reject_rate"] = false_rejects / max(ok_support, 1)
```

```
    metrics["threshold"] = float(threshold)
    return metrics

def compute_metric_confidence_intervals(y_true: np.ndarray, defect_prob: np.
 ↪ndarray, threshold: float) -> pd.DataFrame:
    y_pred = (defect_prob >= threshold).astype(int)
    defect_mask = y_true == TRAINING_CONFIG.defect_positive_label
    ok_mask = y_true != TRAINING_CONFIG.defect_positive_label
    tp = int((defect_mask & (y_pred == 1)).sum())
    fn = int((defect_mask & (y_pred == 0)).sum())
    fp = int((ok_mask & (y_pred == 1)).sum())
    tn = int((ok_mask & (y_pred == 0)).sum())
    rows = []
    for metric_name, successes, total in [
        ("defect_recall", tp, tp + fn),
        ("false_accept_rate", fn, tp + fn),
        ("ok_accept_rate", tn, tn + fp),
        ("false_reject_rate", fp, tn + fp),
    ]:
        low, high = wilson_interval(successes, total)
        estimate = successes / total if total else np.nan
        rows.append(
            {
                "metric": metric_name,
                "estimate": estimate,
                "ci95_low": low,
                "ci95_high": high,
                "successes": successes,
                "total": total,
            }
        )
    return pd.DataFrame(rows)

def evaluate_thresholds(y_true: np.ndarray, defect_prob: np.ndarray) -> pd.
 ↪DataFrame:
    rows = []
    thresholds = np.linspace(0.01, 0.99, TRAINING_CONFIG.threshold_grid_points)
    for threshold in thresholds:
        rows.append(compute_advanced_metrics(y_true, defect_prob, threshold))
    return pd.DataFrame(rows)

val_true, val_pred_argmax, val_prob = predict_loader(model, val_loader)
test_true, test_pred_argmax, test_prob = predict_loader(model, test_loader)

val_threshold_df = evaluate_thresholds(val_true, val_prob[:, TRAINING_CONFIG.
 ↪defect_positive_label])
```

```
display(val_threshold_df.sort_values(["false_accepts", "false_rejects",␣
↪"threshold"], ascending=[True, True, True]).head(15))

eligible = val_threshold_df[val_threshold_df["recall_defect"] >=␣
↪TRAINING_CONFIG.defect_recall_target]
if len(eligible):
    selected = eligible.sort_values(["false_accepts", "false_rejects",␣
↪"threshold"], ascending=[True, True, True]).iloc[0]
else:
    selected = val_threshold_df.sort_values(["recall_defect", "f1_defect"],␣
↪ascending=[False, False]).iloc[0]

OPERATING_THRESHOLD = float(selected["threshold"])
threshold_policy = {
    "positive_class": "def_front",
    "objective": "minimize defective parts accepted as OK while controlling OK␣
↪parts sent to review",
    "defect_recall_target": TRAINING_CONFIG.defect_recall_target,
    "selected_threshold": OPERATING_THRESHOLD,
    "selected_from_split": "validation",
    "selection_sort_order": ["false_accepts ascending", "false_rejects␣
↪ascending", "threshold ascending for conservative defect-recall margin"],
}
print(json.dumps(threshold_policy, indent=2))
RUN_MANIFEST["threshold_policy"] = json_safe(threshold_policy)

val_metrics = compute_advanced_metrics(val_true, val_prob[:, TRAINING_CONFIG.
↪defect_positive_label], OPERATING_THRESHOLD)
test_metrics = compute_advanced_metrics(test_true, test_prob[:, TRAINING_CONFIG.
↪defect_positive_label], OPERATING_THRESHOLD)
metrics_summary_df = pd.DataFrame(
    [
        {"split": "validation", **val_metrics},
        {"split": "test", **test_metrics},
    ]
)
display(metrics_summary_df)

val_ci_df = compute_metric_confidence_intervals(val_true, val_prob[:,␣
↪TRAINING_CONFIG.defect_positive_label], OPERATING_THRESHOLD).
↪assign(split="validation")
test_ci_df = compute_metric_confidence_intervals(test_true, test_prob[:,␣
↪TRAINING_CONFIG.defect_positive_label], OPERATING_THRESHOLD).
↪assign(split="test")
```

```
metric_confidence_df = pd.concat([val_ci_df, test_ci_df],␣
 ↪ignore_index=True)[["split", "metric", "estimate", "ci95_low", "ci95_high",␣
 ↪"successes", "total"]]
display(metric_confidence_df)
RUN_MANIFEST["metric_confidence_intervals"] = metric_confidence_df.
 ↪to_dict(orient="records")

print("Validation report")
print(classification_report(val_true, (val_prob[:, 1] >= OPERATING_THRESHOLD).
 ↪astype(int), target_names=[DISPLAY_NAMES[0], DISPLAY_NAMES[1]],␣
 ↪zero_division=0))
display(confusion_matrix_df(val_true, (val_prob[:, 1] >= OPERATING_THRESHOLD).
 ↪astype(int)))
print("Test report")
print(classification_report(test_true, (test_prob[:, 1] >= OPERATING_THRESHOLD).
 ↪astype(int), target_names=[DISPLAY_NAMES[0], DISPLAY_NAMES[1]],␣
 ↪zero_division=0))
display(confusion_matrix_df(test_true, (test_prob[:, 1] >= OPERATING_THRESHOLD).
 ↪astype(int)))
print(
    "Inspection framing: false accepts are defective parts predicted OK; "
    "false rejects are OK parts routed to defective/review."
)
```

```
    accuracy  precision_defect  recall_defect  f1_defect  roc_auc  pr_auc  \
2   1.000000          1.000000       1.000000   1.000000      1.0     1.0
3   1.000000          1.000000       1.000000   1.000000      1.0     1.0
1   0.997990          0.996466       1.000000   0.998230      1.0     1.0
0   0.994975          0.991213       1.000000   0.995587      1.0     1.0
4   0.998995          1.000000       0.998227   0.999113      1.0     1.0
5   0.998995          1.000000       0.998227   0.999113      1.0     1.0
6   0.998995          1.000000       0.998227   0.999113      1.0     1.0
7   0.998995          1.000000       0.998227   0.999113      1.0     1.0
8   0.998995          1.000000       0.998227   0.999113      1.0     1.0
9   0.998995          1.000000       0.998227   0.999113      1.0     1.0
10  0.998995          1.000000       0.998227   0.999113      1.0     1.0
11  0.998995          1.000000       0.998227   0.999113      1.0     1.0
12  0.998995          1.000000       0.998227   0.999113      1.0     1.0
13  0.998995          1.000000       0.998227   0.999113      1.0     1.0
14  0.998995          1.000000       0.998227   0.999113      1.0     1.0

    brier_score  false_accepts  false_rejects  defect_support  ok_support  \
2      0.001754              0              0             564         431
3      0.001754              0              0             564         431
1      0.001754              0              2             564         431
0      0.001754              0              5             564         431
4      0.001754              1              0             564         431
```

```
5      0.001754              1              0             564         431
6      0.001754              1              0             564         431
7      0.001754              1              0             564         431
8      0.001754              1              0             564         431
9      0.001754              1              0             564         431
10     0.001754              1              0             564         431
11     0.001754              1              0             564         431
12     0.001754              1              0             564         431
13     0.001754              1              0             564         431
14     0.001754              1              0             564         431

    false_accept_rate  false_reject_rate  threshold
2            0.000000           0.000000   0.031778
3            0.000000           0.000000   0.042667
1            0.000000           0.004640   0.020889
0            0.000000           0.011601   0.010000
4            0.001773           0.000000   0.053556
5            0.001773           0.000000   0.064444
6            0.001773           0.000000   0.075333
7            0.001773           0.000000   0.086222
8            0.001773           0.000000   0.097111
9            0.001773           0.000000   0.108000
10           0.001773           0.000000   0.118889
11           0.001773           0.000000   0.129778
12           0.001773           0.000000   0.140667
13           0.001773           0.000000   0.151556
14           0.001773           0.000000   0.162444
```


```
{
  "positive_class": "def_front",
  "objective": "minimize defective parts accepted as OK while controlling OK
parts sent to review",
  "defect_recall_target": 0.95,
  "selected_threshold": 0.03177777777777778,
  "selected_from_split": "validation",
  "selection_sort_order": [
    "false_accepts ascending",
    "false_rejects ascending",
    "threshold ascending for conservative defect-recall margin"
  ]
}
```

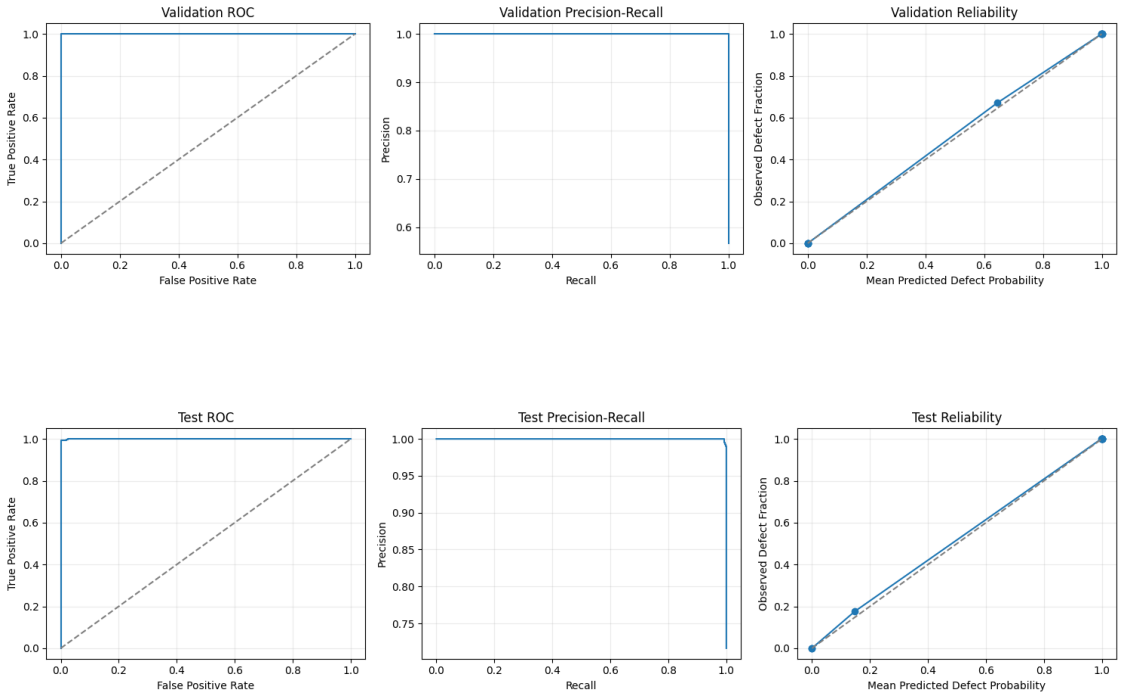

```
        split  accuracy  precision_defect  recall_defect  f1_defect   roc_auc  \
0  validation  1.000000          1.000000            1.0   1.000000  1.000000
1        test  0.992995          0.990315            1.0   0.995134  0.999864

     pr_auc  brier_score  false_accepts  false_rejects  defect_support  \
0  1.000000     0.001754              0              0             564
1  0.999947     0.004337              0              4             409
```

```
   ok_support  false_accept_rate  false_reject_rate  threshold
0         431                0.0           0.000000   0.031778
1         162                0.0           0.024691   0.031778

        split             metric  estimate  ci95_low  ci95_high  successes  \
0  validation      defect_recall  1.000000  0.993235   1.000000        564
1  validation  false_accept_rate  0.000000  0.000000   0.006765          0
2  validation     ok_accept_rate  1.000000  0.991166   1.000000        431
3  validation  false_reject_rate  0.000000  0.000000   0.008834          0
4        test      defect_recall  1.000000  0.990695   1.000000        409
5        test  false_accept_rate  0.000000  0.000000   0.009305          0
6        test     ok_accept_rate  0.975309  0.938240   0.990357        158
7        test  false_reject_rate  0.024691  0.009643   0.061760          4

   total
0    564
1    564
2    431
3    431
4    409
5    409
6    162
7    162

Validation report
              precision    recall  f1-score   support

          OK       1.00      1.00      1.00       431
   Defective       1.00      1.00      1.00       564

    accuracy                           1.00       995
   macro avg       1.00      1.00      1.00       995
weighted avg       1.00      1.00      1.00       995


                  Predicted OK  Predicted Defective
Actual OK                  431                    0
Actual Defective             0                  564

Test report
              precision    recall  f1-score   support

          OK       1.00      0.98      0.99       162
   Defective       0.99      1.00      1.00       409

    accuracy                           0.99       571
   macro avg       1.00      0.99      0.99       571
weighted avg       0.99      0.99      0.99       571
```

```
                   Predicted OK  Predicted Defective
Actual OK                   158                    4
Actual Defective              0                  409

Inspection framing: false accepts are defective parts predicted OK; false
rejects are OK parts routed to defective/review.
```

[17]:
```python
# Code Intent:
# Visualize ranking and calibration behavior beyond scalar metrics.
# Key Ideas:
# - ROC shows separability across thresholds but can be optimistic under␣
↪imbalance.
# - Precision-recall is more informative when defect/OK proportions are uneven.
# - Reliability curves show whether predicted defect probabilities are␣
↪calibrated enough for threshold governance.
# - Plot validation and test curves separately to reveal overfitting or split␣
↪shift.

# ROC, precision-recall, and reliability curve data

def plot_metric_curves(y_true: np.ndarray, defect_prob: np.ndarray,␣
↪title_prefix: str) -> None:
    fig, axes = plt.subplots(1, 3, figsize=(15, 4))

    if len(np.unique(y_true)) > 1:
        fpr, tpr, _ = roc_curve(y_true, defect_prob)
        precision, recall, _ = precision_recall_curve(y_true, defect_prob)
        axes[0].plot(fpr, tpr)
        axes[0].plot([0, 1], [0, 1], linestyle="--", color="gray")
        axes[0].set_title(f"{title_prefix} ROC")
        axes[0].set_xlabel("False Positive Rate")
        axes[0].set_ylabel("True Positive Rate")
        axes[1].plot(recall, precision)
        axes[1].set_title(f"{title_prefix} Precision-Recall")
        axes[1].set_xlabel("Recall")
        axes[1].set_ylabel("Precision")

    prob_true, prob_pred = calibration_curve(y_true, defect_prob, n_bins=10,␣
↪strategy="quantile")
    axes[2].plot(prob_pred, prob_true, marker="o")
    axes[2].plot([0, 1], [0, 1], linestyle="--", color="gray")
    axes[2].set_title(f"{title_prefix} Reliability")
    axes[2].set_xlabel("Mean Predicted Defect Probability")
    axes[2].set_ylabel("Observed Defect Fraction")

    for ax in axes:
        ax.grid(alpha=0.25)
```

```
    plt.tight_layout()
    plt.show()

plot_metric_curves(val_true, val_prob[:, TRAINING_CONFIG.
 ↪defect_positive_label], "Validation")
plot_metric_curves(test_true, test_prob[:, TRAINING_CONFIG.
 ↪defect_positive_label], "Test")
```



[18]:
```
# Code Intent:
# Turn model errors and borderline decisions into reviewable␣
 ↪quality-engineering evidence.
# Key Ideas:
# - Build a row-level prediction table with probabilities, decisions, margins,␣
 ↪and error type.
# - Prioritize false accepts, false rejects, and borderline correct decisions␣
 ↪because they drive release risk and review load.
# - Visual grids help reviewers distinguish real model failures from label␣
 ↪noise or acquisition artifacts.
# - Keep paths in the table so images can be traced back to source data.

# False-accept / false-reject / borderline review tables and visual grids

def build_prediction_frame(df: pd.DataFrame, y_true: np.ndarray, defect_prob:␣
 ↪np.ndarray, threshold: float) -> pd.DataFrame:
```

```
    pred_df = df.reset_index(drop=True).copy()
    pred_df["true_label"] = y_true
    pred_df["defect_probability"] = defect_prob
    pred_df["pred_label"] = (defect_prob >= threshold).astype(int)
    pred_df["pred_display"] = pred_df["pred_label"].map(DISPLAY_NAMES)
    pred_df["error_type"] = "correct"
    pred_df.loc[(pred_df["true_label"] == 1) & (pred_df["pred_label"] == 0),␣
↪"error_type"] = "false_accept"
    pred_df.loc[(pred_df["true_label"] == 0) & (pred_df["pred_label"] == 1),␣
↪"error_type"] = "false_reject"
    pred_df["decision_margin"] = (pred_df["defect_probability"] - threshold).
↪abs()
    pred_df["review_priority"] = 0.0
    pred_df.loc[pred_df["error_type"] == "false_accept", "review_priority"] = 1.
↪0 - pred_df["defect_probability"]
    pred_df.loc[pred_df["error_type"] == "false_reject", "review_priority"] =␣
↪pred_df["defect_probability"]
    pred_df.loc[pred_df["error_type"] == "correct", "review_priority"] = 1.0 -␣
↪np.minimum(pred_df["decision_margin"] / max(threshold, 1e-6), 1.0)
    return pred_df

def plot_review_grid_from_frame(subset: pd.DataFrame, title: str, n: int = 12)␣
↪-> None:
    subset = subset.head(n)
    if subset.empty:
        print(f"No examples available for {title}.")
        return
    cols = min(4, len(subset))
    rows = int(math.ceil(len(subset) / cols))
    fig, axes = plt.subplots(rows, cols, figsize=(cols * 3.0, rows * 3.0))
    axes = np.array(axes).reshape(-1)
    fig.suptitle(title, fontsize=14)
    for ax, (_, row) in zip(axes, subset.iterrows()):
        ax.imshow(read_rgb(row["path"]), cmap="gray")
        ax.set_title(
            f"p_def={row['defect_probability']:.3f}␣
↪margin={row['decision_margin']:.3f}\n"
            f"true={row['display_label']} pred={row['pred_display']}",
            fontsize=9,
        )
        ax.axis("off")
    for ax in axes[len(subset):]:
        ax.axis("off")
    plt.tight_layout()
    plt.show()
```

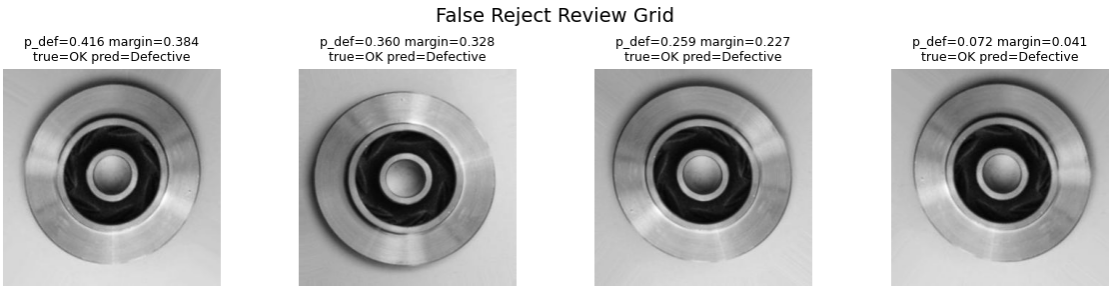
```
def plot_review_grid(pred_df: pd.DataFrame, error_type: str, n: int = 12) ->␣
↪None:
    subset = pred_df[pred_df["error_type"] == error_type].
↪sort_values("review_priority", ascending=False)
    plot_review_grid_from_frame(subset, f"{error_type.replace('_', ' ').
↪title()} Review Grid", n=n)

def plot_borderline_correct_grid(pred_df: pd.DataFrame, n: int = 8) -> None:
    subset = pred_df[pred_df["error_type"] == "correct"].
↪sort_values("decision_margin", ascending=True)
    plot_review_grid_from_frame(subset, "Borderline Correct Review Grid", n=n)

val_prediction_df = build_prediction_frame(val_df, val_true, val_prob[:, 1],␣
↪OPERATING_THRESHOLD)
test_prediction_df = build_prediction_frame(test_df, test_true, test_prob[:,␣
↪1], OPERATING_THRESHOLD)

review_columns = [
    "path",
    "file_name",
    "display_label",
    "defect_probability",
    "pred_display",
    "error_type",
    "decision_margin",
    "review_priority",
]
review_summary_df = test_prediction_df.groupby("error_type").size().
↪reset_index(name="count")
display(review_summary_df)
error_review_df = test_prediction_df[test_prediction_df["error_type"] !=␣
↪"correct"].sort_values("review_priority", ascending=False)
display(error_review_df[review_columns].head(25))

borderline_correct_df = test_prediction_df[test_prediction_df["error_type"] ==␣
↪"correct"].sort_values("decision_margin", ascending=True).head(12)
display(borderline_correct_df[review_columns])
plot_review_grid(test_prediction_df, "false_accept", n=8)
plot_review_grid(test_prediction_df, "false_reject", n=8)
plot_borderline_correct_grid(test_prediction_df, n=8)
```

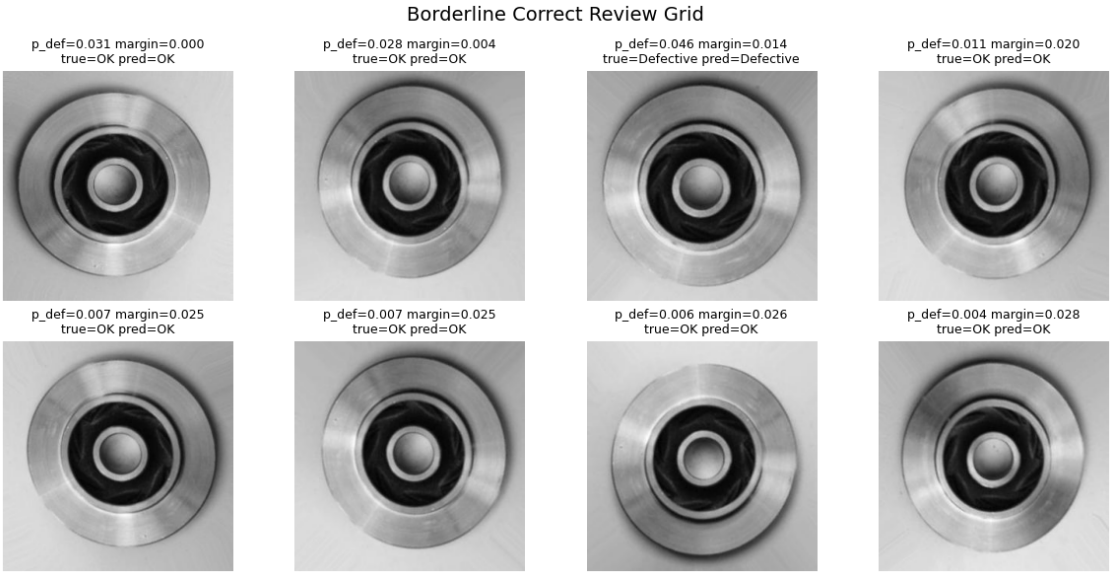
```
     error_type  count
0       correct    567
1  false_reject      4

                                                  path            file_name  \
17   /kaggle/input/real-life-industrial-dataset-of-…  cast_ok_0_1368.jpeg
```

```
135  /kaggle/input/real-life-industrial-dataset-of-…  cast_ok_0_7839.jpeg
123  /kaggle/input/real-life-industrial-dataset-of-…    cast_ok_0_62.jpeg
6    /kaggle/input/real-life-industrial-dataset-of-…  cast_ok_0_1316.jpeg

    display_label  defect_probability pred_display    error_type  \
17             OK            0.415996    Defective  false_reject
135            OK            0.359795    Defective  false_reject
123            OK            0.258671    Defective  false_reject
6              OK            0.072284    Defective  false_reject

     decision_margin  review_priority
17          0.384218         0.415996
135         0.328017         0.359795
123         0.226893         0.258671
6           0.040506         0.072284

                                                   path             file_name  \
43   /kaggle/input/real-life-industrial-dataset-of-…    cast_ok_0_259.jpeg
93   /kaggle/input/real-life-industrial-dataset-of-…   cast_ok_0_2911.jpeg
279  /kaggle/input/real-life-industrial-dataset-of-…  cast_def_0_1591.jpeg
27   /kaggle/input/real-life-industrial-dataset-of-…   cast_ok_0_1426.jpeg
153  /kaggle/input/real-life-industrial-dataset-of-…   cast_ok_0_9926.jpeg
87   /kaggle/input/real-life-industrial-dataset-of-…   cast_ok_0_2869.jpeg
20   /kaggle/input/real-life-industrial-dataset-of-…   cast_ok_0_1382.jpeg
76   /kaggle/input/real-life-industrial-dataset-of-…   cast_ok_0_2810.jpeg
143  /kaggle/input/real-life-industrial-dataset-of-…   cast_ok_0_8010.jpeg
125  /kaggle/input/real-life-industrial-dataset-of-…   cast_ok_0_6401.jpeg
58   /kaggle/input/real-life-industrial-dataset-of-…   cast_ok_0_2730.jpeg
103  /kaggle/input/real-life-industrial-dataset-of-…   cast_ok_0_3292.jpeg

    display_label  defect_probability pred_display error_type  \
43             OK            0.031445           OK    correct
93             OK            0.028176           OK    correct
279     Defective            0.045880    Defective    correct
27             OK            0.011362           OK    correct
153            OK            0.007109           OK    correct
87             OK            0.007066           OK    correct
20             OK            0.006012           OK    correct
76             OK            0.004130           OK    correct
143            OK            0.002886           OK    correct
125            OK            0.002658           OK    correct
58             OK            0.002496           OK    correct
103            OK            0.002432           OK    correct

     decision_margin  review_priority
43          0.000332         0.989542
93          0.003602         0.886663
279         0.014102         0.556237
```

```
27          0.020416        0.357554
153         0.024669        0.223708
87          0.024711        0.222368
20          0.025765        0.189201
76          0.027647        0.129974
143         0.028892        0.090813
125         0.029119        0.083653
58          0.029282        0.078544
103         0.029345        0.076543
```

No examples available for False Accept Review Grid.





1.5.1 Explainability Diagnostics

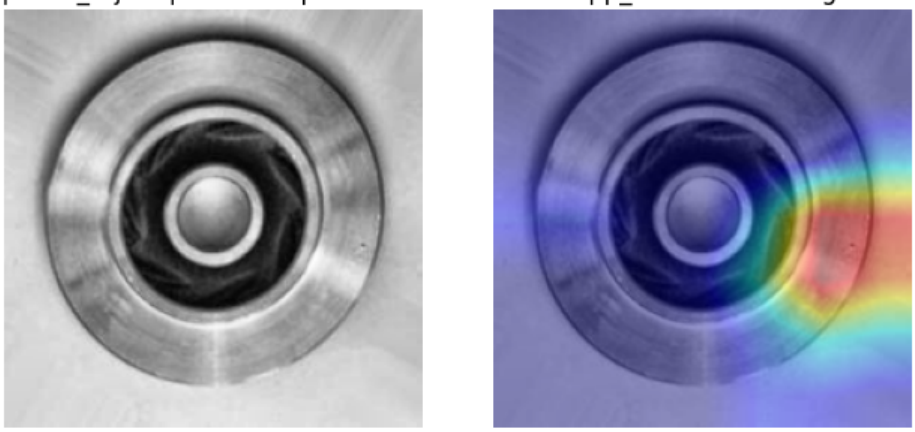
Grad-CAM overlays can help reviewers see where the classifier is most sensitive. These overlays are diagnostic only. They do not prove causality, do not replace localized annotation, and should not be used as final defect evidence without human review.

[19]:
```
# Code Intent:
# Generate diagnostic heatmaps showing image regions that influence the ResNet␣
↪defect score.
# Key Ideas:
# - Hook the final ResNet convolutional block to capture activations and␣
↪gradients.
# - Weight activations by class-specific gradients to form a defect-class␣
↪heatmap.
# - Overlay heatmaps on preprocessed images for reviewer inspection.
# - Prefer errors first, then high-risk borderline correct decisions if no␣
↪errors are available.
# - Treat Grad-CAM as diagnostic support only, not as a substitute for␣
↪localized labels or human evidence.
# - Remove hooks after plotting to avoid stale references in later notebook␣
↪execution.

# Lightweight ResNet Grad-CAM diagnostics

class ResNetGradCAM:
    def __init__(self, model: nn.Module, target_layer: nn.Module):
        self.model = model
        self.target_layer = target_layer
        self.activations = None
        self.gradients = None
        self.forward_handle = target_layer.register_forward_hook(self.
↪_forward_hook)
        self.backward_handle = target_layer.register_full_backward_hook(self.
↪_backward_hook)

    def _forward_hook(self, module, inputs, output):
        self.activations = output.detach()

    def _backward_hook(self, module, grad_input, grad_output):
        self.gradients = grad_output[0].detach()

    def remove(self) -> None:
        self.forward_handle.remove()
        self.backward_handle.remove()

    def __call__(self, input_tensor: torch.Tensor, target_class: int) -> np.
↪ndarray:
        self.model.zero_grad(set_to_none=True)
        logits = self.model(input_tensor)
        score = logits[:, target_class].sum()
        score.backward()
        weights = self.gradients.mean(dim=(2, 3), keepdim=True)
```

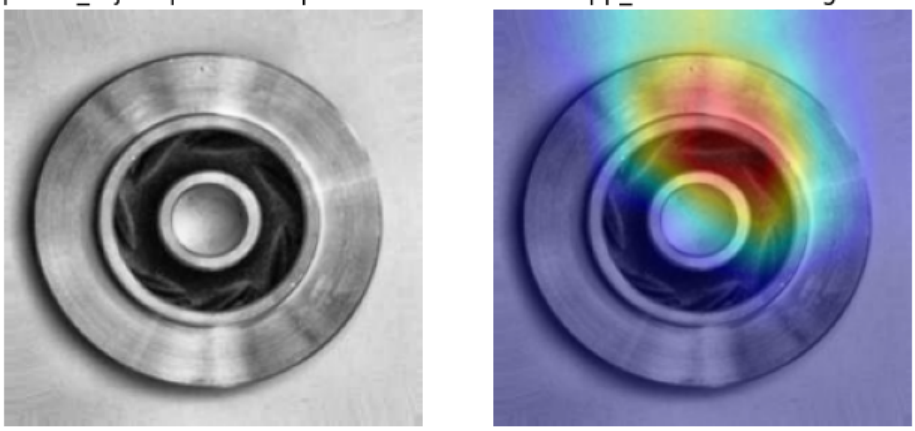
```
        cam = (weights * self.activations).sum(dim=1, keepdim=True)
        cam = F.relu(cam)
        cam = F.interpolate(cam, size=input_tensor.shape[-2:], mode="bilinear",␣
↪align_corners=False)
        heatmap = cam.squeeze().detach().cpu().numpy()
        heatmap = heatmap - heatmap.min()
        heatmap = heatmap / (heatmap.max() + 1e-8)
        return heatmap

def tensor_for_path(path: str) -> torch.Tensor:
    image_bgr = cv2.imread(path, cv2.IMREAD_COLOR)
    image_rgb = preprocess_for_inspection(image_bgr)
    return eval_transform(Image.fromarray(image_rgb)).unsqueeze(0).to(DEVICE)

def overlay_heatmap(image_rgb: np.ndarray, heatmap: np.ndarray, alpha: float =␣
↪0.4) -> np.ndarray:
    resized_heatmap = cv2.resize(heatmap, (image_rgb.shape[1], image_rgb.
↪shape[0]))
    color_heatmap = cv2.applyColorMap(np.uint8(255 * resized_heatmap), cv2.
↪COLORMAP_JET)
    color_heatmap = cv2.cvtColor(color_heatmap, cv2.COLOR_BGR2RGB)
    overlay = np.clip((1 - alpha) * image_rgb + alpha * color_heatmap, 0, 255).
↪astype(np.uint8)
    return overlay

def select_gradcam_review_examples(pred_df: pd.DataFrame, n: int = 6) -> pd.
↪DataFrame:
    errors = pred_df[pred_df["error_type"] != "correct"].
↪sort_values("review_priority", ascending=False)
    if len(errors):
        return errors.head(n)
    return pred_df.sort_values("decision_margin", ascending=True).head(n)

def plot_gradcam_examples(pred_df: pd.DataFrame, n: int = 6) -> None:
    review_df = select_gradcam_review_examples(pred_df, n=n)
    if review_df.empty:
        print("No examples available for Grad-CAM.")
        return

    gradcam = ResNetGradCAM(model.eval(), model.layer4)
    fig, axes = plt.subplots(len(review_df), 2, figsize=(7, 3.2 *␣
↪len(review_df)))
    axes = np.array(axes).reshape(len(review_df), 2)
    try:
        for row_idx, (_, row) in enumerate(review_df.iterrows()):
```

```
            image_rgb = preprocess_for_inspection(cv2.imread(row["path"], cv2.
↪IMREAD_COLOR))
            tensor = tensor_for_path(row["path"])
            heatmap = gradcam(tensor, target_class=TRAINING_CONFIG.
↪defect_positive_label)
            axes[row_idx, 0].imshow(image_rgb)
            axes[row_idx, 0].set_title(f"Input | {row['error_type']} |␣
↪true={row['display_label']} pred={row['pred_display']}")
            axes[row_idx, 0].axis("off")
            axes[row_idx, 1].imshow(overlay_heatmap(image_rgb, heatmap))
            axes[row_idx, 1].set_title(f"Grad-CAM |␣
↪p_def={row['defect_probability']:.3f} margin={row['decision_margin']:.3f}")
            axes[row_idx, 1].axis("off")
    finally:
        gradcam.remove()
    plt.tight_layout()
    plt.show()

plot_gradcam_examples(test_prediction_df, n=4)
```

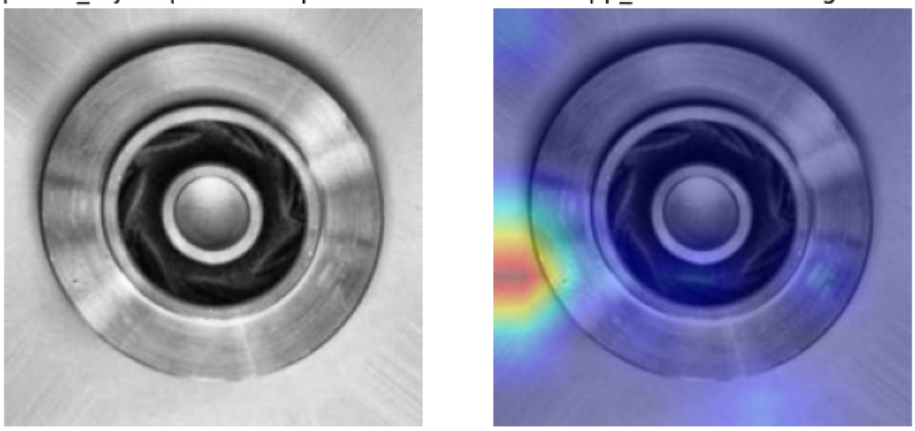
Input | false_reject | true=OK pred=Defective Grad-CAM | p_def=0.416 margin=0.384



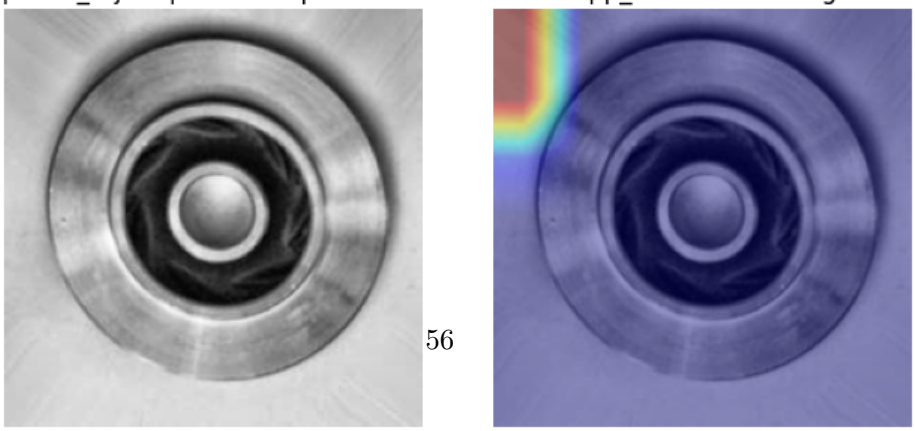
Input | false_reject | true=OK pred=Defective Grad-CAM | p_def=0.360 margin=0.328



Input | false_reject | true=OK pred=Defective Grad-CAM | p_def=0.259 margin=0.227



Input | false_reject | true=OK pred=Defective Grad-CAM | p_def=0.072 margin=0.041



1.6 6. Ultralytics Integration Path

The current dataset is classification-only. Detection, segmentation, and OBB workflows require localized annotation first. The code below defines label contracts and validation helpers so later annotation conversion remains explicit and auditable.

[20]:
```
# Code Intent:
# Define the future Ultralytics localization contract without pretending labels␣
↪already exist.
# Key Ideas:
# - Keep base model names configurable through environment variables for␣
↪YOLOv8+ workflows.
# - Validate expected YOLO image/label folder structure before training␣
↪localized models.
# - Validate detection rows, segmentation polygons, and OBB corners for␣
↪normalized coordinates and known classes.
# - Separate annotation governance from classifier training because image-level␣
↪labels cannot create boxes or masks.

# YOLO dataset contract and label validators

ULTRALYTICS_BASE_MODEL = os.getenv("ULTRALYTICS_BASE_MODEL", "yolov8n.pt")
ULTRALYTICS_SEG_MODEL = os.getenv("ULTRALYTICS_SEG_MODEL", "yolov8n-seg.pt")
ULTRALYTICS_OBB_MODEL = os.getenv("ULTRALYTICS_OBB_MODEL", "yolov8n-obb.pt")

def write_simple_yaml(payload: Dict[str, object], output_path: Path) -> None:
    lines = []
    for key, value in payload.items():
        if isinstance(value, dict):
            lines.append(f"{key}:")
            for inner_key, inner_value in value.items():
                lines.append(f"  {inner_key}: {inner_value}")
        else:
            lines.append(f"{key}: {value}")
    output_path.parent.mkdir(parents=True, exist_ok=True)
    output_path.write_text("\n".join(lines) + "\n", encoding="utf-8")

def build_yolo_dataset_yaml(dataset_root: Path, class_names: Dict[int, str],␣
↪output_path: Path) -> Dict[str, object]:
    payload = {"path": str(dataset_root), "train": "images/train", "val":␣
↪"images/val", "test": "images/test", "names": class_names}
    write_simple_yaml(payload, output_path)
    return payload
```

```python
def validate_yolo_dataset_contract(dataset_root: Path) -> Dict[str, bool]:
    required = {
        "images_train": dataset_root / "images" / "train",
        "images_val": dataset_root / "images" / "val",
        "labels_train": dataset_root / "labels" / "train",
        "labels_val": dataset_root / "labels" / "val",
    }
    return {name: path.exists() for name, path in required.items()}

def assert_normalized(values: Iterable[float], name: str) -> None:
    bad_values = [float(value) for value in values if not (0.0 <= float(value)␣
↪<= 1.0)]
    if bad_values:
        raise ValueError(f"{name} contains values outside [0, 1]: {bad_values[:
↪5]}")

def validate_detection_row(row: Tuple[int, float, float, float, float]) -> None:
    class_id, x_center, y_center, width, height = row
    if class_id not in DEFECT_CLASSES:
        raise ValueError(f"Unknown class id: {class_id}")
    assert_normalized([x_center, y_center, width, height], "Detection row")
    if width <= 0 or height <= 0:
        raise ValueError("Detection width and height must be positive.")

def validate_segmentation_row(row: Tuple[int, Sequence[Tuple[float, float]]])␣
↪-> None:
    class_id, polygon = row
    if class_id not in DEFECT_CLASSES:
        raise ValueError(f"Unknown class id: {class_id}")
    if len(polygon) < 3:
        raise ValueError("Segmentation polygon requires at least 3 points.")
    assert_normalized([coord for xy in polygon for coord in xy], "Segmentation␣
↪polygon")

def validate_obb_row(row: Tuple[int, Sequence[Tuple[float, float]]]) -> None:
    class_id, corners = row
    if class_id not in DEFECT_CLASSES:
        raise ValueError(f"Unknown class id: {class_id}")
    if len(corners) != 4:
        raise ValueError("OBB row requires exactly 4 corners.")
    assert_normalized([coord for xy in corners for coord in xy], "OBB corners")

example_contract = validate_yolo_dataset_contract(Path("datasets/
↪casting_yolo_detect"))
display(pd.DataFrame([example_contract]))
```

```
   images_train  images_val  labels_train  labels_val
0         False       False         False       False
```

[21]:
```
# Code Intent:
# Provide controlled command templates for future YOLO detection, segmentation,␣
↪OBB, and export work.
# Key Ideas:
# - Keep commands visible as reviewable deployment recipes rather than␣
↪executing them implicitly.
# - Use the configured model names so teams can swap approved Ultralytics␣
↪variants.
# - Include an INT8 TensorRT export example tied to a calibration dataset YAML.
# - Make the path from annotation to edge export explicit for later production␣
↪extension.

# Ultralytics train/export command matrix

ultralytics_commands = {
    "detect_train": f"yolo detect train model={ULTRALYTICS_BASE_MODEL}␣
↪data=datasets/casting_yolo_detect/data.yaml imgsz=640 epochs=100 batch=16",
    "segment_train": f"yolo segment train model={ULTRALYTICS_SEG_MODEL}␣
↪data=datasets/casting_yolo_segment/data.yaml imgsz=640 epochs=100 batch=16",
    "obb_train": f"yolo obb train model={ULTRALYTICS_OBB_MODEL} data=datasets/
↪casting_yolo_obb/data.yaml imgsz=640 epochs=100 batch=16",
    "detect_export_tensorrt_int8": "yolo export model=runs/detect/train/weights/
↪best.pt format=engine int8=True data=datasets/casting_yolo_detect/data.yaml",
}

display(pd.DataFrame([{"operation": key, "command": value} for key, value in␣
↪ultralytics_commands.items()]))
```

```
                    operation  \
0                 detect_train
1                segment_train
2                    obb_train
3  detect_export_tensorrt_int8

                                            command
0  yolo detect train model=yolov8n.pt data=datase…
1  yolo segment train model=yolov8n-seg.pt data=d…
2  yolo obb train model=yolov8n-obb.pt data=datas…
3  yolo export model=runs/detect/train/weights/be…
```

1.7 7. Edge Optimization Considerations

Edge readiness requires measuring the full station path: acquisition, transfer, validation, preprocessing, inference, thresholding, postprocessing, logging, and station I/O. Batch-size-1 p95 latency is the governing number.

INT8, TensorRT, OpenVINO, or ONNX Runtime should be selected only after parity and defect-recall checks pass on representative plant imagery.

[22]:
```
# Code Intent:
# Measure edge-relevant single-image inference behavior.
# Key Ideas:
# - Define a single-image prediction wrapper that includes preprocessing and␣
 ↪thresholding.
# - Benchmark batch-size-1 pipeline latency because station decisions happen␣
 ↪one part at a time.
# - Report p95 latency against a configurable budget rather than relying on␣
 ↪average latency.
# - Synchronize CUDA timing when applicable so GPU timings are meaningful.

# Batch-size-1 edge latency benchmark

@torch.no_grad()
def predict_single_image(path: str, threshold: float = OPERATING_THRESHOLD) ->␣
 ↪Dict[str, object]:
    model.eval()
    image_bgr = cv2.imread(path, cv2.IMREAD_COLOR)
    image_rgb = preprocess_for_inspection(image_bgr)
    tensor = eval_transform(Image.fromarray(image_rgb)).unsqueeze(0).to(DEVICE)
    start = time.perf_counter()
    logits = model(tensor)
    if DEVICE.type == "cuda":
        torch.cuda.synchronize()
    model_latency_ms = (time.perf_counter() - start) * 1000
    probabilities = torch.softmax(logits, dim=1).cpu().numpy()[0]
    defect_probability = float(probabilities[TRAINING_CONFIG.
 ↪defect_positive_label])
    predicted_label = int(defect_probability >= threshold)
    return {
        "path": path,
        "predicted_label": DISPLAY_NAMES[predicted_label],
        "defect_probability": defect_probability,
        "threshold": threshold,
        "latency_ms_model_only": model_latency_ms,
    }

@torch.no_grad()
def benchmark_single_image_pipeline(model: nn.Module, sample_paths:␣
 ↪Sequence[str]) -> pd.DataFrame:
    model.eval()
    paths = list(sample_paths)
    if not paths:
        raise ValueError("No sample paths provided.")
```

```
    for i in range(EDGE_CONFIG.benchmark_warmup):
        _ = predict_single_image(paths[i % len(paths)],␣
 ↪threshold=OPERATING_THRESHOLD)
    timings = []
    for i in range(EDGE_CONFIG.benchmark_repeats):
        path = paths[i % len(paths)]
        start = time.perf_counter()
        _ = predict_single_image(path, threshold=OPERATING_THRESHOLD)
        timings.append((time.perf_counter() - start) * 1000)
    return pd.DataFrame({"pipeline_latency_ms": timings})

latency_sample_paths = test_df.sample(min(16, len(test_df)),␣
 ↪random_state=SEED)["path"].tolist()
latency_df = benchmark_single_image_pipeline(model, latency_sample_paths)
latency_summary = latency_df.describe(percentiles=[0.5, 0.9, 0.95, 0.99])
latency_p95_ms = float(latency_df["pipeline_latency_ms"].quantile(0.95))
latency_budget_passed = latency_p95_ms <= EDGE_CONFIG.p95_latency_budget_ms
display(latency_summary)
print(f"p95 latency: {latency_p95_ms:.2f} ms | budget: {EDGE_CONFIG.
 ↪p95_latency_budget_ms:.2f} ms | passed={latency_budget_passed}")
```

```
       pipeline_latency_ms
count            50.000000
mean              7.499226
std               0.402024
min               6.953262
50%               7.507948
90%               7.808921
95%               8.266615
99%               8.727941
max               8.765647

p95 latency: 8.27 ms | budget: 50.00 ms | passed=True
```

[23]:
```
# Code Intent:
# Package the model decision system only when explicitly enabled.
# Key Ideas:
# - Keep artifact saving off by default to prevent accidental filesystem churn␣
 ↪during review.
# - Save model weights, threshold policy, label map, manifest, and metrics␣
 ↪together for traceability.
# - Export ONNX only when requested and when optional dependencies are␣
 ↪available.
# - Track actual artifact and parity status instead of inferring success from␣
 ↪configuration flags.

# Artifact save, ONNX export, and optional ONNX Runtime parity check
```

```python
def write_json(path: Path, payload: Dict[str, object]) -> None:
    path.parent.mkdir(parents=True, exist_ok=True)
    path.write_text(json.dumps(json_safe(payload), indent=2), encoding="utf-8")

def build_artifact_status() -> Dict[str, object]:
    return {
        "artifact_root": str(ARTIFACT_CONFIG.root),
        "save_requested": bool(ARTIFACT_CONFIG.save_artifacts),
        "export_onnx_requested": bool(ARTIFACT_CONFIG.export_onnx),
        "onnx_parity_requested": bool(ARTIFACT_CONFIG.validate_onnx_parity),
        "model_state_saved": False,
        "metadata_saved": False,
        "onnx_exported": False,
        "onnx_parity_checked": False,
        "onnx_parity_passed": False,
        "onnx_max_abs_diff": None,
        "notes": [],
    }

artifact_status = build_artifact_status()
artifact_payload = {
    "manifest": RUN_MANIFEST,
    "threshold_policy": threshold_policy,
    "label_map": INDEX_TO_CLASS,
    "metrics": metrics_summary_df.to_dict(orient="records"),
    "metric_confidence_intervals": metric_confidence_df.
 ↪to_dict(orient="records"),
    "latency": {
        "p95_ms": latency_p95_ms,
        "budget_ms": EDGE_CONFIG.p95_latency_budget_ms,
        "passed": latency_budget_passed,
    },
}

if ARTIFACT_CONFIG.save_artifacts:
    ARTIFACT_CONFIG.root.mkdir(parents=True, exist_ok=True)
    torch.save(model.state_dict(), ARTIFACT_CONFIG.root / "model.pt")
    write_json(ARTIFACT_CONFIG.root / "threshold.json", threshold_policy)
    write_json(ARTIFACT_CONFIG.root / "label_map.json", INDEX_TO_CLASS)
    write_json(ARTIFACT_CONFIG.root / "manifest.json", RUN_MANIFEST)
    write_json(ARTIFACT_CONFIG.root / "artifact_payload.json", artifact_payload)
    metrics_summary_df.to_csv(ARTIFACT_CONFIG.root / "metrics.csv", index=False)
    artifact_status["model_state_saved"] = True
    artifact_status["metadata_saved"] = True
    artifact_status["notes"].append("Model and metadata artifacts saved.")
    print(f"Saved model artifacts under {ARTIFACT_CONFIG.root.resolve()}")
else:
```

```
    artifact_status["notes"].append("Artifact saving disabled by configuration.
↪")
    print("Artifact saving disabled. Set SAVE_CASTING_ARTIFACTS=true to write␣
↪model.pt, threshold.json, label_map.json, manifest.json, artifact_payload.
↪json, and metrics.csv.")

ONNX_PATH = ARTIFACT_CONFIG.root / "casting_resnet18_classifier.onnx"
if ARTIFACT_CONFIG.export_onnx:
    if importlib.util.find_spec("onnx") is None:
        artifact_status["notes"].append("ONNX export skipped because the␣
↪optional 'onnx' package is not installed.")
        print("ONNX export skipped because the optional 'onnx' package is not␣
↪installed.")
    else:
        ARTIFACT_CONFIG.root.mkdir(parents=True, exist_ok=True)
        model.eval()
        dummy_input = torch.randn(1, 3, TRAINING_CONFIG.image_size,␣
↪TRAINING_CONFIG.image_size, device=DEVICE)
        torch.onnx.export(
            model,
            dummy_input,
            str(ONNX_PATH),
            input_names=["image"],
            output_names=["logits"],
            dynamic_axes=None,
            opset_version=17,
        )
        artifact_status["onnx_exported"] = True
        artifact_status["notes"].append(f"Exported ONNX model: {ONNX_PATH.
↪resolve()}")
        print(f"Exported ONNX model: {ONNX_PATH.resolve()}")

        if ARTIFACT_CONFIG.validate_onnx_parity and importlib.util.
↪find_spec("onnxruntime") is not None:
            import onnxruntime as ort

            session = ort.InferenceSession(str(ONNX_PATH),␣
↪providers=["CPUExecutionProvider"])
            with torch.no_grad():
                pt_logits = model(dummy_input).detach().cpu().numpy()
            ort_logits = session.run(None, {"image": dummy_input.detach().cpu().
↪numpy()})[0]
            max_abs_diff = float(np.max(np.abs(pt_logits - ort_logits)))
            artifact_status["onnx_parity_checked"] = True
            artifact_status["onnx_max_abs_diff"] = max_abs_diff
            artifact_status["onnx_parity_passed"] = max_abs_diff <= 1e-4
```

```
            print(f"ONNX parity max_abs_diff={max_abs_diff:.6f} |␣
↪passed={artifact_status['onnx_parity_passed']}")
        elif ARTIFACT_CONFIG.validate_onnx_parity:
            artifact_status["notes"].append("ONNX parity skipped because␣
↪optional 'onnxruntime' is not installed.")
            print("ONNX parity skipped because optional 'onnxruntime' is not␣
↪installed.")
else:
    artifact_status["notes"].append("ONNX export disabled by configuration.")
    print(f"ONNX export disabled. Set EXPORT_CASTING_ONNX=true to export␣
↪{ONNX_PATH}.")

RUN_MANIFEST["artifact_status"] = json_safe(artifact_status)
display(pd.DataFrame([artifact_status]).drop(columns=["notes"]))
print(json.dumps({"artifact_notes": artifact_status["notes"]}, indent=2))
```

```
Artifact saving disabled. Set SAVE_CASTING_ARTIFACTS=true to write model.pt,
threshold.json, label_map.json, manifest.json, artifact_payload.json, and
metrics.csv.
ONNX export disabled. Set EXPORT_CASTING_ONNX=true to export
artifacts/casting_inspection/casting_resnet18_classifier.onnx.

                  artifact_root  save_requested  export_onnx_requested  \
0  artifacts/casting_inspection           False                  False

   onnx_parity_requested  model_state_saved  metadata_saved  onnx_exported  \
0                   True              False           False          False

   onnx_parity_checked  onnx_parity_passed onnx_max_abs_diff
0                False               False              None

{
  "artifact_notes": [
    "Artifact saving disabled by configuration.",
    "ONNX export disabled by configuration."
  ]
}
```

1.8 8. Retraining & Drift Handling Strategy

Production retraining should be driven by station telemetry, human review, downstream escapes, and drift indicators rather than fixed calendar retraining alone.

Version together: dataset snapshot, label schema, preprocessing code, training code, model weights, threshold, export artifact, calibration data, validation report, and rollback artifact.

[24]:
```
# Code Intent:
# Specify what production inference logs should contain and how review queues␣
↪can be prioritized.
```

```python
# Key Ideas:
# - Include station metadata, model version, preprocessing version, threshold,␣
↪probability, decision, diagnostics, and latency.
# - Build one example log record so the schema is concrete and testable.
# - Score review priority by uncertainty around the threshold because␣
↪borderline cases are most informative.
# - Keep operator override fields in the schema because human-in-the-loop␣
↪feedback is part of retraining.

# Production inference log schema and review prioritization

INFERENCE_LOG_COLUMNS = [
    "event_id",
    "timestamp_utc",
    "plant_id",
    "line_id",
    "station_id",
    "camera_id",
    "part_number",
    "model_version",
    "preprocess_version",
    "threshold",
    "defect_probability",
    "decision",
    "operator_override",
    "override_reason",
    "image_uri",
    "mean_intensity",
    "std_intensity",
    "edge_density",
    "laplacian_variance",
    "latency_ms",
]

def build_inference_log_record(
    image_path: str,
    prediction: Dict[str, object],
    station_metadata: Dict[str, str],
) -> Dict[str, object]:
    image_bgr = cv2.imread(image_path, cv2.IMREAD_COLOR)
    diagnostics = texture_diagnostics(image_bgr) if image_bgr is not None else␣
↪{}
    return {
        "event_id": hashlib.sha1(f"{image_path}-{time.time()}".encode("utf-8")).
↪hexdigest()[:16],
        "timestamp_utc": datetime.now(timezone.utc).isoformat(),
        "model_version": station_metadata.get("model_version", MODEL_VERSION),
```

```
        "preprocess_version": station_metadata.get("preprocess_version",␣
↪PREPROCESS_VERSION),
        "threshold": prediction.get("threshold"),
        "defect_probability": prediction.get("defect_probability"),
        "decision": prediction.get("predicted_label"),
        "operator_override": None,
        "override_reason": None,
        "image_uri": image_path,
        "latency_ms": prediction.get("latency_ms_model_only"),
        **{key: station_metadata.get(key) for key in ["plant_id", "line_id",␣
↪"station_id", "camera_id", "part_number"]},
        **diagnostics,
    }

def review_priority_score(record: Dict[str, object]) -> float:
    probability = float(record.get("defect_probability", 0.5))
    threshold = float(record.get("threshold", OPERATING_THRESHOLD))
    uncertainty = 1.0 - min(abs(probability - threshold) / max(threshold,␣
↪1e-6), 1.0)
    return float(uncertainty)

example_prediction = predict_single_image(test_df.iloc[0]["path"],␣
↪threshold=OPERATING_THRESHOLD)
example_log_record = build_inference_log_record(
    test_df.iloc[0]["path"],
    example_prediction,
    {
        "plant_id": "pilot_plant",
        "line_id": "casting_line_01",
        "station_id": "vision_station_01",
        "camera_id": "top_view_01",
        "part_number": "public_impeller_reference",
    },
)
example_log_record["review_priority"] =␣
↪review_priority_score(example_log_record)
display(pd.DataFrame([example_log_record], columns=INFERENCE_LOG_COLUMNS +␣
↪["review_priority"]))
```

```
          event_id                      timestamp_utc     plant_id  \
0  bb053fc8a972a227  2026-05-01T02:26:32.243093+00:00  pilot_plant

          line_id         station_id    camera_id                part_number  \
0  casting_line_01  vision_station_01  top_view_01  public_impeller_reference

                         model_version preprocess_version  threshold  …  \
0  resnet18-imagelevel-public-pilot-v1    opencv-clahe-v2   0.031778  …
```

```
   decision operator_override override_reason  \
0        OK               None            None

                                          image_uri mean_intensity  \
0  /kaggle/input/real-life-industrial-dataset-of-…     138.967456

   std_intensity  edge_density  laplacian_variance  latency_ms  \
0      61.914225      0.045156          277.378987    4.020802

   review_priority
0         0.000113

[1 rows x 21 columns]
```

[25]:
```
# Code Intent:
# Provide lightweight dataset drift monitoring using image-quality features.
# Key Ideas:
# - Extract intensity, contrast, edge density, and sharpness diagnostics from␣
↪sampled final split images.
# - Use Population Stability Index to compare current data against a final␣
↪training reference distribution.
# - Interpret PSI with review bands so the number becomes an alerting signal,␣
↪not an automatic retraining decision.
# - Display feature distributions by class to help separate visual drift from␣
↪label mix changes.

# Drift monitoring features and PSI checks

def extract_monitoring_features(df: pd.DataFrame, sample_size: Optional[int] =␣
↪500) -> pd.DataFrame:
    working_df = df.copy()
    if sample_size is not None:
        working_df = working_df.sample(min(sample_size, len(working_df)),␣
↪random_state=SEED)
    rows = []
    for _, row in working_df.iterrows():
        image_bgr = cv2.imread(row["path"], cv2.IMREAD_COLOR)
        if image_bgr is None:
            continue
        rows.append(
            {
                "path": row["path"],
                "split": row.get("model_split", row.get("split", "unknown")),
                "label": row["label"],
                "display_label": row["display_label"],
                **texture_diagnostics(image_bgr),
```

```
            }
        )
    return pd.DataFrame(rows)

def population_stability_index(
    expected: np.ndarray,
    actual: np.ndarray,
    bins: int = 10,
    epsilon: float = 1e-6,
) -> float:
    breakpoints = np.unique(np.quantile(expected, np.linspace(0, 1, bins + 1)))
    if len(breakpoints) < 3:
        return 0.0
    expected_counts, _ = np.histogram(expected, bins=breakpoints)
    actual_counts, _ = np.histogram(actual, bins=breakpoints)
    expected_pct = np.maximum(expected_counts / max(expected_counts.sum(), 1),␣
 ↪epsilon)
    actual_pct = np.maximum(actual_counts / max(actual_counts.sum(), 1),␣
 ↪epsilon)
    return float(np.sum((actual_pct - expected_pct) * np.log(actual_pct /␣
 ↪expected_pct)))

def interpret_psi(value: float) -> str:
    if value < 0.10:
        return "stable"
    if value < 0.25:
        return "monitor"
    return "investigate"

train_monitor = extract_monitoring_features(train_df, sample_size=500)
test_monitor = extract_monitoring_features(test_df, sample_size=500)
psi_rows = []
for feature in ["mean_intensity", "std_intensity", "edge_density",␣
 ↪"laplacian_variance"]:
    psi_value = population_stability_index(train_monitor[feature].values,␣
 ↪test_monitor[feature].values)
    psi_rows.append(
        {
            "feature": feature,
            "psi_train_vs_test": psi_value,
            "interpretation": interpret_psi(psi_value),
        }
    )
psi_df = pd.DataFrame(psi_rows)
display(psi_df)
display(train_monitor.groupby("display_label")[["mean_intensity",␣
 ↪"std_intensity", "edge_density", "laplacian_variance"]].describe())
```

```
RUN_MANIFEST["drift_monitoring"] = psi_df.to_dict(orient="records")
```

```
              feature  psi_train_vs_test interpretation
0      mean_intensity           0.162970        monitor
1       std_intensity           0.193809        monitor
2        edge_density           0.130976        monitor
3  laplacian_variance           0.203481        monitor

              mean_intensity                                                \
                       count        mean       std         min         25%
display_label
Defective              274.0  139.726610  6.180578  124.910411  135.360219
OK                     226.0  149.651886  6.248698  127.603744  143.056919

                                                   std_intensity             \
                     50%         75%         max         count       mean
display_label
Defective      139.432267  143.079622  156.586589         274.0  58.860906
OK             151.660494  153.679567  159.256844         226.0  62.596573

               … edge_density           laplacian_variance              \
               …          75%       max              count        mean
display_label  …
Defective      …     0.049092  0.057244              274.0  185.869516
OK             …     0.043086  0.053511              226.0  170.077442

                                                                        \
                      std        min         25%         50%         75%
display_label
Defective      60.730994  69.970788  140.340282  184.082905  231.300026
OK             75.152165  69.973788   91.345119  163.243938  223.139424


                     max
display_label
Defective      328.173255
OK             435.900751

[2 rows x 32 columns]
```

1.9 9. Risks & Mitigations

1.9.1 Rare Defect Under-Representation

Risk: rare but high-cost defect modes may be absent. Mitigate with rare-defect review sets, downstream escape sampling, active learning, and localized labels for recurring high-cost defects. Low-data methods can be evaluated later as secondary support, not as an unvalidated production replacement.

1.9.2 Overfitting to Visual Artifacts

Risk: the model may learn lighting, fixture, background, camera, compression, or augmentation artifacts. Mitigate with station-specific holdouts, hard negatives, image diagnostics, and false-decision review.

1.9.3 Plant-to-Plant Variability

Risk: model performance may degrade across plants, suppliers, part variants, or camera stations. Mitigate with plant-level acceptance metrics, metadata capture, staged rollout, local calibration sets, and site-level rollback.

1.9.4 Edge Deployment Failures

Risk: thermal throttling, camera faults, runtime mismatch, corrupt frames, or latency spikes. Mitigate with watchdogs, fail-safe routing, runtime versioning, health checks, and tested rollback.

1.10 10. Acceptance Criteria

1.10.1 Technical Acceptance

- Dataset integrity checks pass.
- Preprocessing is deterministic and documented.
- Defect recall meets the agreed target on plant-specific holdout data.
- False accepts are reviewed and dispositioned.
- Threshold selection is risk-based and versioned.
- Batch-size-1 p95 latency meets station budget.
- Exported artifact matches PyTorch baseline within tolerance.
- Monitoring features and inference logs are available.

1.10.2 Operational Acceptance

- Operators can challenge and override decisions.
- Review workflow exists for uncertain or disputed images.
- Model version and threshold are visible in station logs.
- Rollback procedure is tested.
- Maintenance has a documented recovery path.
- Plant quality engineering signs off on defect definitions and sampling.

[26]:
```
# Code Intent:
# Convert acceptance criteria into an auditable checklist and final␣
↪interview-facing executive summary.
# Key Ideas:
# - Combine data integrity, final leakage status, metric reporting, threshold␣
↪selection, latency, artifacts, and operational readiness.
# - Mark plant-specific holdout and exported artifact validation explicitly␣
↪false unless actually completed.
# - Compute checklist values from notebook state rather than hard-coding pass␣
↪values.
```

```
# - State the release decision as pilot/reference-ready only, not␣
↪production-approved.

# Acceptance checklist, release readiness, and executive summary

def build_release_readiness_summary() -> pd.DataFrame:
    final_filename_clean = leakage_summary["final_filename_leak_pairs"] == 0
    final_hash_clean = leakage_summary["final_hash_leak_pairs"] == 0
    rows = [
        {
            "criterion": "data_integrity_passed",
            "passed": audit_summary["unreadable_images"] == 0,
            "evidence": f"{audit_summary['scanned_images']} scanned,␣
↪{audit_summary['unreadable_images']} unreadable",
        },
        {
            "criterion": "no_filename_leakage",
            "passed": final_filename_clean,
            "evidence": f"final filename overlap␣
↪count={leakage_summary['final_filename_leak_pairs']}",
        },
        {
            "criterion": "no_exact_hash_leakage",
            "passed": final_hash_clean,
            "evidence": f"final SHA-1 overlap␣
↪count={leakage_summary['final_hash_leak_pairs']}",
        },
        {
            "criterion": "preprocessing_contract_passed",
            "passed": "validate_preprocessing" in globals(),
            "evidence": f"{PREPROCESS_VERSION}, {TRAINING_CONFIG.
↪image_size}x{TRAINING_CONFIG.image_size} RGB uint8 before tensor␣
↪normalization",
        },
        {
            "criterion": "advanced_metrics_reported",
            "passed": {"roc_auc", "pr_auc", "brier_score", "false_accept_rate",␣
↪"false_reject_rate"}.issubset(metrics_summary_df.columns),
            "evidence": "ROC-AUC, PR-AUC, Brier, false-accept, false-reject␣
↪metrics computed",
        },
        {
            "criterion": "threshold_selected_for_defect_recall",
            "passed": "OPERATING_THRESHOLD" in globals() and␣
↪val_metrics["recall_defect"] >= TRAINING_CONFIG.defect_recall_target,
```

```
            "evidence": f"threshold={OPERATING_THRESHOLD:.6f}, validation␣
↪recall={val_metrics['recall_defect']:.6f}",
        },
        {
            "criterion": "test_defect_recall_meets_target",
            "passed": bool(test_metrics["recall_defect"] >= TRAINING_CONFIG.
↪defect_recall_target),
            "evidence": f"test recall={test_metrics['recall_defect']:.6f},␣
↪target={TRAINING_CONFIG.defect_recall_target:.2f}",
        },
        {
            "criterion": "batch_size_one_latency_measured",
            "passed": "latency_p95_ms" in globals(),
            "evidence": f"p95={latency_p95_ms:.2f} ms",
        },
        {
            "criterion": "latency_budget_passed",
            "passed": bool(latency_budget_passed),
            "evidence": f"budget={EDGE_CONFIG.p95_latency_budget_ms:.2f} ms",
        },
        {
            "criterion": "plant_specific_holdout_validated",
            "passed": False,
            "evidence": "Not available in public dataset; required before␣
↪production release",
        },
        {
            "criterion": "exported_artifact_validated",
            "passed": bool(artifact_status.get("onnx_parity_passed")),
            "evidence": "ONNX/export parity only passes when export and parity␣
↪check are actually executed",
        },
        {
            "criterion": "rollback_plan_documented",
            "passed": True,
            "evidence": "Model card and release gates require versioned␣
↪rollback artifact",
        },
        {
            "criterion": "operator_review_workflow_defined",
            "passed": "review_priority" in test_prediction_df.columns and␣
↪"operator_override" in INFERENCE_LOG_COLUMNS,
            "evidence": "Review priority and operator override fields are␣
↪present",
        },
    ]
```

```
    return pd.DataFrame(rows)

release_readiness_summary = build_release_readiness_summary()
acceptance_criteria = dict(zip(release_readiness_summary["criterion"],␣
 ↪release_readiness_summary["passed"]))

pilot_reference_gate_names = [
    "data_integrity_passed",
    "no_filename_leakage",
    "no_exact_hash_leakage",
    "preprocessing_contract_passed",
    "advanced_metrics_reported",
    "threshold_selected_for_defect_recall",
    "test_defect_recall_meets_target",
    "batch_size_one_latency_measured",
    "latency_budget_passed",
    "operator_review_workflow_defined",
]
pilot_reference_ready = all(bool(acceptance_criteria[name]) for name in␣
 ↪pilot_reference_gate_names)
computed_notebook_release_status =␣
 ↪"pilot_reference_ready_not_production_approved" if pilot_reference_ready␣
 ↪else "pilot_reference_has_blockers"
production_approved = all(bool(value) for value in acceptance_criteria.values())

executive_summary_df = pd.DataFrame(
    [
        {"area": "Release decision", "value": computed_notebook_release_status},
        {"area": "Production approval", "value": bool(production_approved)},
        {"area": "Final test images", "value": int(len(test_df))},
        {"area": "Test defect recall", "value":␣
 ↪f"{test_metrics['recall_defect']:.6f}"},
        {"area": "Test false accepts", "value":␣
 ↪int(test_metrics["false_accepts"])},
        {"area": "Test false rejects", "value":␣
 ↪int(test_metrics["false_rejects"])},
        {"area": "Operating threshold", "value": f"{OPERATING_THRESHOLD:.6f}"},
        {"area": "p95 batch-size-1 latency", "value": f"{latency_p95_ms:.2f}␣
 ↪ms"},
        {"area": "Final leakage removals", "value":␣
 ↪int(len(leakage_removals_df))},
        {"area": "Plant holdout validated", "value": False},
        {"area": "Exported artifact parity", "value": bool(artifact_status.
 ↪get("onnx_parity_passed"))},
    ]
)
```

```
display(executive_summary_df)
display(release_readiness_summary)
RUN_MANIFEST["notebook_release_status"] = computed_notebook_release_status
RUN_MANIFEST["production_approved"] = bool(production_approved)
RUN_MANIFEST["release_readiness_summary"] = release_readiness_summary.
 ↪to_dict(orient="records")
RUN_MANIFEST["executive_summary"] = executive_summary_df.
 ↪to_dict(orient="records")
```

```
                       area                                          value
0          Release decision  pilot_reference_ready_not_production_approved
1       Production approval                                          False
2         Final test images                                            571
3        Test defect recall                                       1.000000
4        Test false accepts                                              0
5        Test false rejects                                              4
6       Operating threshold                                       0.031778
7   p95 batch-size-1 latency                                       8.27 ms
8     Final leakage removals                                           144
9    Plant holdout validated                                         False
10  Exported artifact parity                                         False

                             criterion  passed  \
0                 data_integrity_passed    True
1                   no_filename_leakage    True
2                 no_exact_hash_leakage    True
3         preprocessing_contract_passed    True
4             advanced_metrics_reported    True
5   threshold_selected_for_defect_recall    True
6        test_defect_recall_meets_target    True
7        batch_size_one_latency_measured    True
8                 latency_budget_passed    True
9       plant_specific_holdout_validated   False
10          exported_artifact_validated   False
11             rollback_plan_documented    True
12      operator_review_workflow_defined    True

                                          evidence
0                        7348 scanned, 0 unreadable
1                     final filename overlap count=0
2                        final SHA-1 overlap count=0
3   opencv-clahe-v2, 224x224 RGB uint8 before tens…
4   ROC-AUC, PR-AUC, Brier, false-accept, false-re…
5      threshold=0.031778, validation recall=1.000000
6                  test recall=1.000000, target=0.95
7                                       p95=8.27 ms
8                                   budget=50.00 ms
```

```
9   Not available in public dataset; required befo…
10  ONNX/export parity only passes when export and…
11  Model card and release gates require versioned…
12  Review priority and operator override fields a…
```

1.10.3 Model Card: Casting Inspection Baseline

Intended use: part-level OK/defective screening for controlled casting imagery in a pilot inspection workflow.

Non-use: final production release, safety certification, defect localization, severity scoring, supplier chargeback, or rework routing without plant-specific validation and additional localized labels.

Training data: public Kaggle casting inspection dataset with image-level binary labels.

Known limits: possible acquisition artifacts, unknown plant-to-plant generalization, limited defect taxonomy, no bounding boxes or masks, and potential repeated/augmented samples.

Decision policy: threshold selected on validation data to prioritize defective-part recall and reduce false accepts.

Edge target: batch-size-1 inference with deterministic OpenCV preprocessing and explicit fail-safe routing for unreadable or failed inference events.

Release gates: plant-specific holdout pass, false-accept review, latency benchmark pass, artifact parity check, rollback test, operator review workflow, and quality engineering approval.

1.11 11. Test Strategy

1.11.1 Unit-Level Tests

Validate folder structure, class mapping, image readability, preprocessing output shape/range, file leakage, average-hash duplicate risk, and metric calculation.

1.11.2 Model-Level Tests

Validate one-batch overfit behavior, metric reporting, threshold sensitivity, calibration, false accepts, false rejects, and Grad-CAM diagnostic generation.

1.11.3 System-Level Tests

Validate batch-size-1 inference, latency budget, corrupt-frame handling, exported model parity, logging schema, watchdog behavior, artifact versioning, and rollback.

[27]:
```
# Code Intent:
# Run lightweight notebook tests and define safe station behavior for failed␣
↪inference.
# Key Ideas:
# - Re-check dataset structure, final split leakage, metric math,␣
↪preprocessing, threshold policy, and YOLO label validators.
# - Fail-safe prediction routes unreadable or failed images to manual review␣
↪rather than accepting the part.
```

```
# - Return a structured failure payload so station software can log and act on␣
↪failures consistently.
# - Keep this cell near the end so it validates the full set of notebook␣
↪definitions.

# Notebook-level tests and fail-safe inference wrapper

def safe_inspection_predict(path: str, threshold: float = OPERATING_THRESHOLD)␣
↪-> Dict[str, object]:
    """
    Fail-safe station wrapper. If loading or inference fails, route to manual␣
↪review rather than accepting the part.
    """
    try:
        return {"status": "model_decision", **predict_single_image(path,␣
↪threshold=threshold)}
    except Exception as exc:
        return {
            "status": "manual_review_required",
            "path": path,
            "reason": str(exc),
            "predicted_label": "Defective",
            "defect_probability": 1.0,
            "threshold": threshold,
        }

def test_expected_dataset_structure(root: Path) -> None:
    for split in ["train", "test"]:
        for class_name in CLASS_TO_INDEX:
            expected = root / split / class_name
            assert expected.exists(), f"Missing expected folder: {expected}"

def test_no_split_leakage(train_frame: pd.DataFrame, val_frame: pd.DataFrame,␣
↪test_frame: pd.DataFrame) -> None:
    assert_no_filename_leakage(("train", train_frame), ("val", val_frame),␣
↪("test", test_frame))
    assert_no_hash_leakage(("train", train_frame), ("val", val_frame), ("test",␣
↪test_frame))

def test_metric_calculation_contract() -> None:
    y_true_contract = np.array([0, 0, 1, 1])
    y_pred_contract = np.array([0, 1, 1, 1])
    metrics = compute_binary_metrics(y_true_contract, y_pred_contract)
    assert np.isclose(metrics["accuracy"], 0.75)
    assert np.isclose(metrics["precision_defect"], 2 / 3)
    assert np.isclose(metrics["recall_defect"], 1.0)
```

```
    assert np.isclose(metrics["f1_defect"], 0.8)

def test_threshold_policy_contract() -> None:
    assert threshold_policy["selected_from_split"] == "validation"
    assert val_metrics["recall_defect"] >= TRAINING_CONFIG.defect_recall_target
    assert test_metrics["recall_defect"] >= TRAINING_CONFIG.defect_recall_target
    assert test_metrics["false_accept_rate"] <= 1.0 - TRAINING_CONFIG.
 ↪defect_recall_target + 1e-12

def test_prediction_frame_alignment() -> None:
    assert len(test_prediction_df) == len(test_df)
    assert np.array_equal(test_prediction_df["true_label"].values, test_true)
    assert {"decision_margin", "review_priority", "error_type"}.
 ↪issubset(test_prediction_df.columns)

def test_fail_safe_inference_wrapper() -> None:
    failed = safe_inspection_predict("nonexistent_image.png")
    assert failed["status"] == "manual_review_required"
    assert failed["predicted_label"] == "Defective"
    assert failed["defect_probability"] == 1.0

def test_acceptance_checklist_contract() -> None:
    assert acceptance_criteria["no_filename_leakage"]
    assert acceptance_criteria["no_exact_hash_leakage"]
    assert acceptance_criteria["threshold_selected_for_defect_recall"]
    assert acceptance_criteria["test_defect_recall_meets_target"]
    assert acceptance_criteria["plant_specific_holdout_validated"] is False
    assert acceptance_criteria["exported_artifact_validated"] ==␣
 ↪bool(artifact_status.get("onnx_parity_passed"))

def run_notebook_tests() -> None:
    test_expected_dataset_structure(DATA_ROOT)
    test_no_split_leakage(train_df, val_df, test_df)
    test_metric_calculation_contract()
    test_threshold_policy_contract()
    test_prediction_frame_alignment()
    test_fail_safe_inference_wrapper()
    test_acceptance_checklist_contract()
    sampled_paths = dataset_df.sample(min(25, len(dataset_df)),␣
 ↪random_state=SEED)["path"]
    for path in sampled_paths:
        validate_preprocessing(path)
    validate_detection_row((0, 0.5, 0.5, 0.2, 0.2))
    validate_segmentation_row((0, [(0.1, 0.1), (0.2, 0.1), (0.2, 0.2)]))
    validate_obb_row((0, [(0.1, 0.1), (0.2, 0.1), (0.2, 0.2), (0.1, 0.2)]))
    print("Notebook test suite passed.")
```

```
run_notebook_tests()
display(pd.DataFrame([safe_inspection_predict("nonexistent_image.png")]))
```

```
Notebook test suite passed.

                   status                   path                reason  \
0  manual_review_required  nonexistent_image.png  Input image is None

  predicted_label  defect_probability  threshold
0       Defective                 1.0   0.031778
```

1.12 Closing Engineering Notes

This notebook includes dataset integrity controls, reproducibility metadata, final split leakage governance, risk-based thresholding, confidence intervals, calibration review, error and borderline-case review, explainability diagnostics, optional artifact/export governance, edge latency checks, drift strategy, acceptance gates, notebook-level tests, and a model card.

The classifier remains the correct first step because the public dataset is image-level only. YOLO detection, segmentation, and OBB should be introduced only after localized annotations exist and plant quality engineering approves the taxonomy.

The final notebook release decision is intentionally conservative: it can be used as a pilot/reference workflow, but it is not production-approved. Production release still requires plant-specific holdout validation, false-accept review with quality engineering, exported-runtime parity, rollback testing, and operational signoff.

1.12.1 Reference Links

- Kaggle dataset: https://www.kaggle.com/datasets/ravirajsinh45/real-life-industrial-dataset-of-casting-product
- Ultralytics dataset documentation: https://docs.ultralytics.com/datasets/
- Ultralytics export documentation: https://docs.ultralytics.com/modes/export/
- TorchVision ResNet18 documentation: https://docs.pytorch.org/vision/stable/models/generated/torchvision

[28]:
```
import json

# Display the final RUN_MANIFEST for documentation
print(json.dumps(RUN_MANIFEST, indent=2, default=str))
```

```
{
  "workflow": "casting_inspection_industrial_workflow",
  "created_utc": "2026-05-01T02:21:16.467442+00:00",
  "seed": 42,
  "device": "cuda",
  "git_commit": null,
  "model_version": "resnet18-imagelevel-public-pilot-v1",
  "preprocess_version": "opencv-clahe-v2",
  "notebook_release_status": "pilot_reference_ready_not_production_approved",
  "leakage_policy": {
```

```
    "priority_order": [
      "train",
      "val",
      "test"
    ],
    "train_val_overlap_action": "remove_from_validation",
    "test_overlap_action": "remove_from_test",
    "filename_overlap": "drop_downstream_rows_matching_upstream_filenames",
    "exact_hash_overlap": "drop_downstream_rows_matching_upstream_sha1",
    "near_duplicate_overlap": "warn_and_review_only"
  },
  "package_versions": {
    "opencv-python": "4.13.0.92",
    "matplotlib": "3.10.0",
    "numpy": "2.0.2",
    "pandas": "2.2.2",
    "Pillow": "11.3.0",
    "scikit-learn": "1.6.1",
    "torch": "2.10.0+cu128",
    "torchvision": "0.25.0+cu128",
    "onnx": "not-installed",
    "onnxruntime": "not-installed",
    "ultralytics": "not-installed",
    "kagglehub": "1.0.0"
  },
  "configuration": {
    "workflow": {
      "model_version": "resnet18-imagelevel-public-pilot-v1",
      "preprocess_version": "opencv-clahe-v2",
      "notebook_release_status": "pilot_reference_not_production_approved",
      "leakage_policy": {
        "priority_order": [
          "train",
          "val",
          "test"
        ],
        "train_val_overlap_action": "remove_from_validation",
        "test_overlap_action": "remove_from_test",
        "filename_overlap": "drop_downstream_rows_matching_upstream_filenames",
        "exact_hash_overlap": "drop_downstream_rows_matching_upstream_sha1",
        "near_duplicate_overlap": "warn_and_review_only"
      }
    },
    "dataset": {
      "root_env_var": "CASTING_DATA_ROOT",
      "kaggle_dataset_slug": "ravirajsinh45/real-life-industrial-dataset-of-
casting-product",
      "auto_download_dataset": true,
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  "install_kagglehub_if_missing": true,
  "image_extensions": [
    ".jpg",
    ".jpeg",
    ".png",
    ".bmp",
    ".tif",
    ".tiff"
  ],
  "val_fraction_from_train": 0.15,
  "max_audit_images": null,
  "max_near_duplicate_images": 1500,
  "near_duplicate_hamming_threshold": 4,
  "expected_public_counts_enabled": true
},
"training": {
  "image_size": 224,
  "batch_size": 32,
  "num_workers": 0,
  "epochs": 5,
  "learning_rate": 0.0001,
  "weight_decay": 0.0001,
  "use_weighted_sampler": true,
  "use_class_weighted_loss": false,
  "defect_positive_label": 1,
  "defect_recall_target": 0.95,
  "threshold_grid_points": 91,
  "run_one_batch_smoke_test": false
},
"preprocess": {
  "use_clahe": true,
  "denoise": false,
  "clahe_clip_limit": 2.0,
  "clahe_tile_grid_size": [
    8,
    8
  ]
},
"edge": {
  "p95_latency_budget_ms": 50.0,
  "benchmark_repeats": 50,
  "benchmark_warmup": 10,
  "batch_size_one_required": true
},
"artifact": {
  "root": "artifacts/casting_inspection",
  "save_artifacts": false,
  "export_onnx": false,
```

```
      "validate_onnx_parity": true
    }
  },
  "dataset_root": "/kaggle/input/real-life-industrial-dataset-of-casting-
product/casting_data/casting_data",
  "initial_split_counts": [
    {
      "model_split": "test",
      "display_label": "Defective",
      "count": 453
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    {
      "model_split": "test",
      "display_label": "OK",
      "count": 262
    },
    {
      "model_split": "train",
      "display_label": "Defective",
      "count": 3194
    },
    {
      "model_split": "train",
      "display_label": "OK",
      "count": 2444
    },
    {
      "model_split": "val",
      "display_label": "Defective",
      "count": 564
    },
    {
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      "display_label": "OK",
      "count": 431
    }
  ],
  "audit_summary": {
    "scanned_images": 7348,
    "unreadable_images": 0,
    "unique_shape_count": 1
  },
  "pre_policy_exact_hash_leak_groups": 64,
  "near_duplicate_pairs_shown": 25,
  "final_split_counts": [
    {
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      "display_label": "Defective",
```

```
    "count": 409
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  {
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    "count": 162
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    "display_label": "OK",
    "count": 2444
  },
  {
    "model_split": "val",
    "display_label": "Defective",
    "count": 564
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  {
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    "display_label": "OK",
    "count": 431
  }
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    "display_label": "Defective",
    "count": 409
  },
  {
    "model_split": "test",
    "display_label": "OK",
    "count": 162
  },
  {
    "model_split": "train",
    "display_label": "Defective",
    "count": 3194
  },
  {
    "model_split": "train",
    "display_label": "OK",
    "count": 2444
```

```
    },
    {
      "model_split": "val",
      "display_label": "Defective",
      "count": 564
    },
    {
      "model_split": "val",
      "display_label": "OK",
      "count": 431
    }
  ],
  "leakage_summary": {
    "policy": {
      "priority_order": [
        "train",
        "val",
        "test"
      ],
      "train_val_overlap_action": "remove_from_validation",
      "test_overlap_action": "remove_from_test",
      "filename_overlap": "drop_downstream_rows_matching_upstream_filenames",
      "exact_hash_overlap": "drop_downstream_rows_matching_upstream_sha1",
      "near_duplicate_overlap": "warn_and_review_only"
    },
    "removed_from_validation": 0,
    "removed_from_test": 144,
    "final_filename_leak_pairs": 0,
    "final_hash_leak_pairs": 0,
    "near_duplicate_action": "warn_and_review_only"
  },
  "leakage_removal_counts": [
    {
      "removed_from": "test",
      "reason": "filename_overlap",
      "count": 80
    },
    {
      "removed_from": "test",
      "reason": "filename_overlap+exact_hash_overlap",
      "count": 64
    }
  ],
  "threshold_policy": {
    "positive_class": "def_front",
    "objective": "minimize defective parts accepted as OK while controlling OK
parts sent to review",
    "defect_recall_target": 0.95,
```

```
    "selected_threshold": 0.03177777777777778,
    "selected_from_split": "validation",
    "selection_sort_order": [
      "false_accepts ascending",
      "false_rejects ascending",
      "threshold ascending for conservative defect-recall margin"
    ]
  },
  "metric_confidence_intervals": [
    {
      "split": "validation",
      "metric": "defect_recall",
      "estimate": 1.0,
      "ci95_low": 0.9932347330664044,
      "ci95_high": 1.0,
      "successes": 564,
      "total": 564
    },
    {
      "split": "validation",
      "metric": "false_accept_rate",
      "estimate": 0.0,
      "ci95_low": 0.0,
      "ci95_high": 0.006765266933595566,
      "successes": 0,
      "total": 564
    },
    {
      "split": "validation",
      "metric": "ok_accept_rate",
      "estimate": 1.0,
      "ci95_low": 0.9911655186624281,
      "ci95_high": 1.0,
      "successes": 431,
      "total": 431
    },
    {
      "split": "validation",
      "metric": "false_reject_rate",
      "estimate": 0.0,
      "ci95_low": 0.0,
      "ci95_high": 0.008834481337572118,
      "successes": 0,
      "total": 431
    },
    {
      "split": "test",
      "metric": "defect_recall",
```

```
    "estimate": 1.0,
    "ci95_low": 0.9906947361893761,
    "ci95_high": 0.9999999999999998,
    "successes": 409,
    "total": 409
  },
  {
    "split": "test",
    "metric": "false_accept_rate",
    "estimate": 0.0,
    "ci95_low": 0.0,
    "ci95_high": 0.009305263810623734,
    "successes": 0,
    "total": 409
  },
  {
    "split": "test",
    "metric": "ok_accept_rate",
    "estimate": 0.9753086419753086,
    "ci95_low": 0.9382397029988918,
    "ci95_high": 0.9903572232246856,
    "successes": 158,
    "total": 162
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  {
    "split": "test",
    "metric": "false_reject_rate",
    "estimate": 0.024691358024691357,
    "ci95_low": 0.009642776775314481,
    "ci95_high": 0.06176029700110833,
    "successes": 4,
    "total": 162
  }
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"artifact_status": {
  "artifact_root": "artifacts/casting_inspection",
  "save_requested": false,
  "export_onnx_requested": false,
  "onnx_parity_requested": true,
  "model_state_saved": false,
  "metadata_saved": false,
  "onnx_exported": false,
  "onnx_parity_checked": false,
  "onnx_parity_passed": false,
  "onnx_max_abs_diff": null,
  "notes": [
    "Artifact saving disabled by configuration.",
    "ONNX export disabled by configuration."
```

```
    ]
  },
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    {
      "feature": "mean_intensity",
      "psi_train_vs_test": 0.16296956071657057,
      "interpretation": "monitor"
    },
    {
      "feature": "std_intensity",
      "psi_train_vs_test": 0.19380898697081866,
      "interpretation": "monitor"
    },
    {
      "feature": "edge_density",
      "psi_train_vs_test": 0.1309755672566311,
      "interpretation": "monitor"
    },
    {
      "feature": "laplacian_variance",
      "psi_train_vs_test": 0.20348123446175825,
      "interpretation": "monitor"
    }
  ],
  "production_approved": false,
  "release_readiness_summary": [
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      "passed": true,
      "evidence": "7348 scanned, 0 unreadable"
    },
    {
      "criterion": "no_filename_leakage",
      "passed": true,
      "evidence": "final filename overlap count=0"
    },
    {
      "criterion": "no_exact_hash_leakage",
      "passed": true,
      "evidence": "final SHA-1 overlap count=0"
    },
    {
      "criterion": "preprocessing_contract_passed",
      "passed": true,
      "evidence": "opencv-clahe-v2, 224x224 RGB uint8 before tensor
normalization"
    },
    {
```

```
      "criterion": "advanced_metrics_reported",
      "passed": true,
      "evidence": "ROC-AUC, PR-AUC, Brier, false-accept, false-reject metrics
computed"
    },
    {
      "criterion": "threshold_selected_for_defect_recall",
      "passed": true,
      "evidence": "threshold=0.031778, validation recall=1.000000"
    },
    {
      "criterion": "test_defect_recall_meets_target",
      "passed": true,
      "evidence": "test recall=1.000000, target=0.95"
    },
    {
      "criterion": "batch_size_one_latency_measured",
      "passed": true,
      "evidence": "p95=8.27 ms"
    },
    {
      "criterion": "latency_budget_passed",
      "passed": true,
      "evidence": "budget=50.00 ms"
    },
    {
      "criterion": "plant_specific_holdout_validated",
      "passed": false,
      "evidence": "Not available in public dataset; required before production
release"
    },
    {
      "criterion": "exported_artifact_validated",
      "passed": false,
      "evidence": "ONNX/export parity only passes when export and parity check
are actually executed"
    },
    {
      "criterion": "rollback_plan_documented",
      "passed": true,
      "evidence": "Model card and release gates require versioned rollback
artifact"
    },
    {
      "criterion": "operator_review_workflow_defined",
      "passed": true,
      "evidence": "Review priority and operator override fields are present"
    }
```

```
  ],
  "executive_summary": [
    {
      "area": "Release decision",
      "value": "pilot_reference_ready_not_production_approved"
    },
    {
      "area": "Production approval",
      "value": false
    },
    {
      "area": "Final test images",
      "value": 571
    },
    {
      "area": "Test defect recall",
      "value": "1.000000"
    },
    {
      "area": "Test false accepts",
      "value": 0
    },
    {
      "area": "Test false rejects",
      "value": 4
    },
    {
      "area": "Operating threshold",
      "value": "0.031778"
    },
    {
      "area": "p95 batch-size-1 latency",
      "value": "8.27 ms"
    },
    {
      "area": "Final leakage removals",
      "value": 144
    },
    {
      "area": "Plant holdout validated",
      "value": false
    },
    {
      "area": "Exported artifact parity",
      "value": false
    }
  ]
}
```